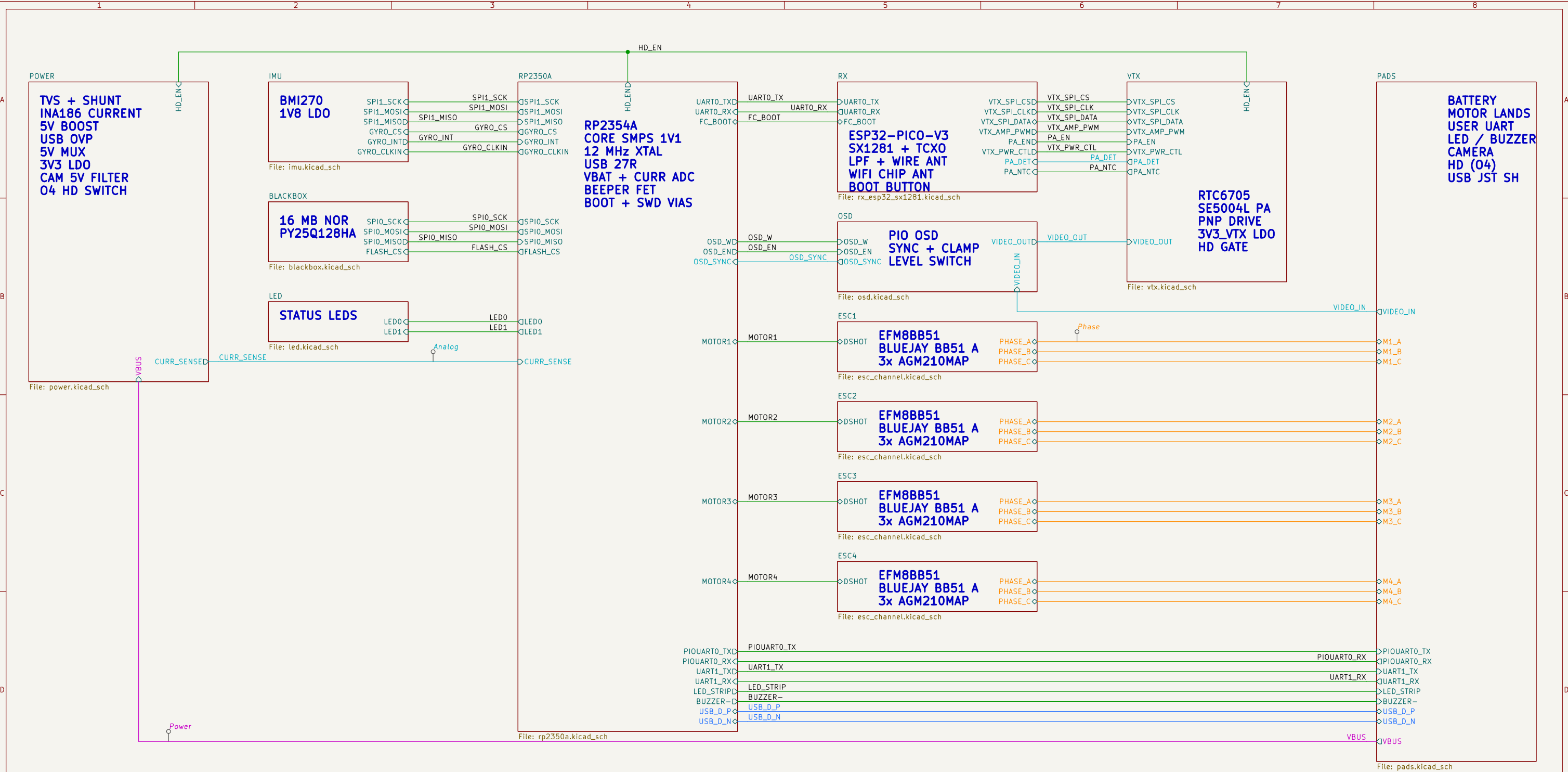




POWER TABLE

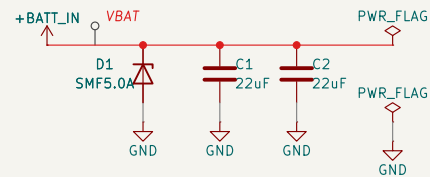
| Rail     | Source                      | Nominal    | Budget (spec 5)                             | Feeds                                  |
|----------|-----------------------------|------------|---------------------------------------------|----------------------------------------|
| +BATT_IN | battery pads J2 / J3 (1S)   | 3.0–4.35 V | 4x ESC + 2.0 A board                        | TVS, pad bulk, shunt R1                |
| +BATT    | shunt R1 0.5 m0hm           | cell       | ESC + boost in 1.4–3.0 A                    | ESC1–4, boost, 04 EN, VBAT ADC, PA_VCC |
| /VBUS    | USB J31 pin 4 (hier. net)   | 5 V        | 0.42 A, USB only                            | U24 OVP                                |
| +5V_USB  | U24 SGM40661 OVP            | 5 V        | as VBUS                                     | U3 VIN1, PR1 divider                   |
| +5V_BST  | U2 TPS61022 boost           | 5.15 V     | 0.70 A analog / 1.48 A HD (BEC 1.5 A cont.) | U3 VIN2, U5 04 switch                  |
| +5V      | U3 TPS2116 mux              | 5 V        | 0.70 A analog, about 0.25 A HD              | 3V3 LDOs, CAM filter, 5V pads          |
| +5V_CAM  | FB1 ferrite from +5V        | 5 V        | 0.12 A                                      | camera pad + plug                      |
| +5V_HD   | U5 TPS22810 (HD_EN, cell)   | 5 V        | 1.20 A 04 Lite                              | VHD pad                                |
| +3V3     | U4 LP5912–3.3               | 3.3 V      | 0.20 A (0.48 A Wi–Fi bench)                 | FC, IMU, NOR, OSD, RX, LEDs, INA186    |
| +3V3_VTX | U21 LP5912–3.3 (off in HD)  | 3.3 V      | 0.10 A typ, about 0.17 A at max PA          | RTC6705, PA drive, PA ref              |
| +1V8     | U12 TPS7A2018               | 1.8 V      | < 1 mA                                      | BMI270                                 |
| +1V1     | RP2354A core SMPS           | 1.1 V      | internal                                    | RP2354A DVDD                           |
| PA_VCC   | R67 selector, +BATT default | cell       | 0.65 A at 27 dBm                            | SE5004L PA (local, VTX)                |
| GND      | battery –                   | 0 V        | –                                           | all sheets                             |

Generated from hardware/sch\_contract.json by hardware/tools/sch\_root.py; do not edit by hand.  
Root wires carry the project-wide net names (root label = net /NAME). Motor phases are named by the ESC sheets (/ESCn/PHASE\_x). Rails are power symbols (table); VBUS is a hierarchical net.  
ESC1–ESC4 drive M1 rear right, M2 front right, M3 rear left, M4 front left (Betaflight quad–X); FC pin map: research/PINMAP.md.

OpenDrone

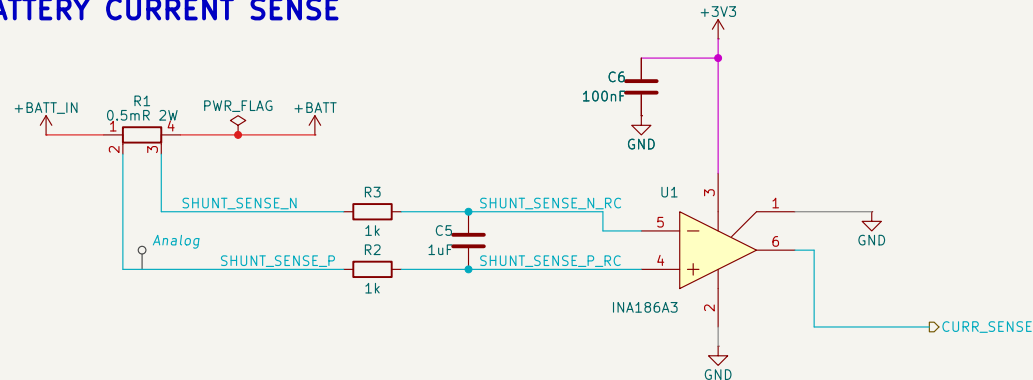
|                                              |                  |           |
|----------------------------------------------|------------------|-----------|
| Sheet: /<br>File: OpenAIO–Whoop.kicad_sch    |                  |           |
| Title: OpenAIO–Whoop overview (1S whoop AIO) |                  |           |
| Size: A3                                     | Date: 2026–10–09 | Rev: rev1 |
| KiCad E.D.A. 10.0.6                          |                  | Id: 1/14  |

## BATTERY INPUT AND TVS



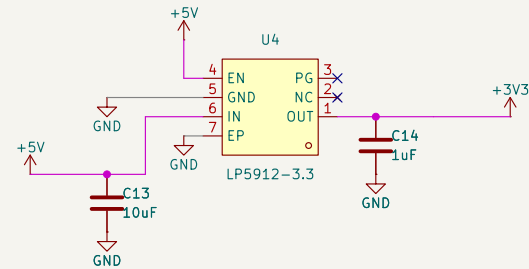
Source: OpenAIO/hardware/OpenAIO/cad\_sch (battery entry), adapted.  
+BAT1\_JIN = battery pads J2 (+) / J3 (-) on the PADS sheet, 1S LiPo/LiHV  
3.0-4.35 V; PWR\_FLAG here (the pads are passive). No reverse-polarity FET (spec  
4.1) - user-fitted pitfall, risk in README.  
D1 SMF5.0A: VBR 6.40-7.07 V, VC 9.2 V at 21.7 A, 200 W 10/1000 us: take one  
more unit's regen energy (-0.29 J per rated pulse) on a battery disconnect; fallback  
SMA5.0A (SMA, 400 W) if V3b exceeds it! It does not hold the 5.5-7 V parts in  
rating (EF8M VDD on the cell; TP561022 VIN 7 V abs max, and its pass-through  
current a surge to +5V\_BST = TP52116 VIN2, 6 V abs max); Matrix-parity risk,  
see 15.2.  
C1/C2 4.5V up 16 V X5R 0603 at the pads; 9-13 uF effective each at 4.35 V. Hot  
plug of a 4.35 V pack can overshoot to 5.0-5.4 V (V3c); above the TP561022 4.8  
V VIN limit for a start-up with VOUT pre-biased < 0.7 V (DS 6.3), which is  
exactly hot plug (EN = VIN, +5V\_BST at 0 V): V3c scopes VIN with the +5V\_BST  
start. Damping fallback -100 uF polymer at the pads; PE-8 waits for V3c.

## BATTERY CURRENT SENSE



Source: OpenAIO/hardware/OpenAIO.kicad\_sch (shunt + INA186A3, root sheet); input RC moved to 0201 and its two common-mode caps deleted (DS2C BR-07); the output RC (1k/100n) sits at the ADC pin (RP2350A sheet, R47/C60).  
P4-9: R1 HC5206 0.5 mOhm on the four-pad Kelvin net-tie land (pad groups 1-2 and 3-4). SHUNT\_SENSE\_PC = pad 2 (battery side).  
SHUNT\_SENSE\_N = pad 3: each holds only its inner sense pad and R2/R3; SHUNT\_SENSE\_PC/N\_RC hold R2/R3, C5 and U1 (netlist checked); all four match the Analog and Kelvin DRU patterns. Route the pair on L1 or L6, never in2/In3 (DRU). class Analog. Everything that draws current taps the R4.  
R2/R3 1k + C5 1 uF differential:  $c = 1 / (2 \pi \times f \times 20 \text{ kOhm} \times 1 \text{ uF}) = 80 \text{ Hz}$ ,  $\text{ILB} < c = 3 \text{ nA} \times 1 \text{ kOhm} = 3 \text{ uV}$  (6 mA),  $\text{VOS} < 50 \text{ uV}$  (0.1 A).  
U1 INA186A3, gain 100: 50 mOhm  $\times 100 = 50 \text{ mV/A}$ , Betaflight ibata\_scale 500: output to VS - 40 mV gives about 65 A full scale on +3V3; VCM -0.2..0.4 V covers the cell; REF to GND (unidirectional); C6 100 nF at VS. Shunt loss 0.2 W at 20 A. Scale checked on every prototype (V-pan).

### 3.3 V LDO FOR FC + RX



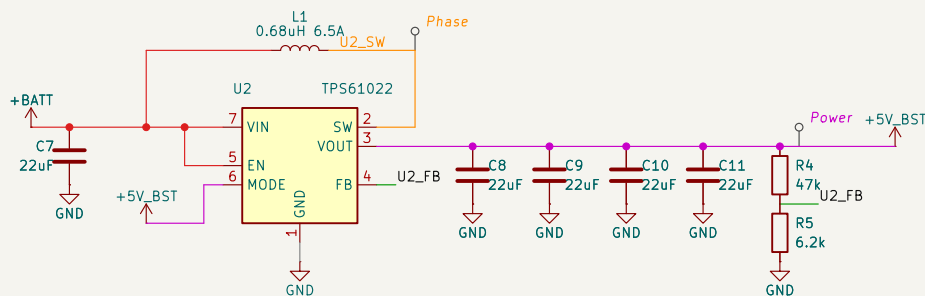
Source: OpenFC-Lite-Mini/hardware/power/kicad.sch (3.3V 0.5A MAIN LDO", LP5912-3.3), with the OpenDrone LP5912-3.3DRVR symbol (pins in package order: IN/EN on the left here, EN tied to +5V, no EN shared with another rail; U21 on VTX has its own EN, PG and NC open; EP to GND for heat (T1: GND or open)).

P4-17 / BR-14: C13 10 uF X6S 0402 is the one CIN of U4 and U21 (VTX sheet): place it between the two IN pins (-2.4 mm apart). 2.0 uF effective at 1.5 V (Murata curve) > the 0.7 uF LP5912 minimum.

C20 12-3V3 0.01 uF at 3.3 V (C82 10 uF 5 V, Murata CWS5C80J106M11 DC-bias curve). C58 4.7 uF 0402 X5R -2.0 V, 3x 1 uF 0201 X5R (C14, C61, C92) 1 uF each, 150 uF of X6S 0201 -0.085 uF each (last three typical, i); 8.7 uF typical, 7.1 uF at -2.0 V, 10.3 uF only with every part at +20 % -> Inside T1: 0.7-10 uF (SNSV470 7.6) but for that +3 % corner, and most of it is remote (5-30 mm of trace), which the LP5912 allows. Lever if a V-bench load step rings: PS7-06 (delete C92). +3V3\_VTX is summed on the VTX sheet.

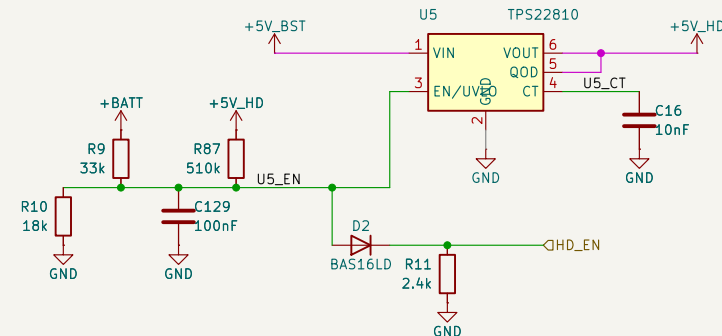
Load 0.20 A: (5.15 - 3.3) V x 0.2 A = 0.37 W, +26 K (RthJA 71.2 K/W). U4 UOT (project-lib LP5912 copy, power=not pin) drives +3V3: no PWR\_FLAG.

## 5 V BOOST TPS61022



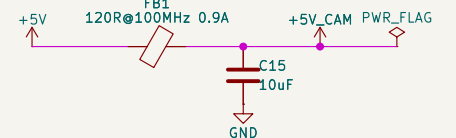
New block from TI: SLVSDXDT7 (grabbing uses the TPS61022); look matched to the OpenFIC-Lite-Mini buck boards.  
P4-7: VOUT = 0.600 V (x1 + 47k / 6.2k) = 5.148 V; with VFB 585=615 mV and 1 % resistors 4.93=5.37 V. R5 6.2k < 300 kOhm (TI). EN tied to VIN (+BATT) and nothing else (BR-11): same -0.3-7 V abs max, VEN\_H <= 1.2 V, on whenever the cell is above UVLO. MODE = VOUT (+5V\_BST): forced PWM, VMODE\_H <= 1.2 V valid once VOUT > 2.2 V (start-up pass-through holds VOUT near VIN).  
C7 22 uF CIN at VIN/GND. C8-C11 4x 22 uF X6S 0603; -20-28 uF effective at 5.15 V, inside TI's 10-50 uF; TI 6.3 asks >= 10 uF effective up to 1.5 A and >= 20 uF from 1.5 to 3 A, met only typically, so the published BEC is 1.5 A continuous (D79; HD load 1.43-1.48 A, LED strip limited in HD mode; V1 load step at 1.7 A). Feed-forward cap (TI 8.2.4; above 40 uF effective, FFZ 2 kHz = 1.7 nF across R4); C8-C11 give 20-28 uF. HD mode adds C12/C13 (-6 uF via the mux) and the O4 input (-47 uF, UFE2 note: none fitted (learn). V1b steps the load in HD with a real O4, 1.5-2.2 nF across R4 only if it rings. L1 0.68 uH, Isat 6.5 A, 17 mOhm; valley limit 6.5-10 A; converter capability 1.67/1.82/1.98 A at 2.8/3.0/3.2 V (spec 4.2).  
Passes VIN to VOUT when VIN > VOUT (surge path, spec 4.1).  
Layout (TI 10.1): Cout VOUT/GND pads at pins 3/1 (hot loop SW-VOUT-Cout-GND smallest). CIN at pins 7/1. SW copper minimal. R4/R5 at FB pin 4 away from SW; ground plane under the part.

## 04 LITE HD SUPPLY SWITCH TPS22810



New block from TI SLVSDH0C (D15, D34; a53 BR-10/22/24).  
P4-2 / BR-10: U5 VIN on +5V\_BST, ahead of the mux: USB never reaches the O4. No separate CIN (C17 deleted, P4 critique): the boost outputs C8-C11 (20-28 uF effective) are U5's input capacitor (TI 10.3: CIN optional, 1 uF usually sufficient); P5 condition: U5 VIN within 3 mm of C8-C11 or U3 VIN2 on the +5V\_BST copper. C16 CT 10 nF: <4.7 V/ms, 0.2 A inrush per 47 uF. QOD tied to VOUT: the internal RPLD empties +5V\_HD when off.  
P4-1: R9 35k / R10 18k from +BATT, R87 510k from +5V\_HD, C129 100 nF, D2 anode and U5 pin 3 (netlist checked). D2 BAS16LD cathode + R11 2.4k pull-down on HD\_EN (GPIO27, PINIO1).  
P4-1 (VENV 1.13-1.30 V, VENF 1.08-1.18 V, +5V\_HD 4.93-5.37 V): EN = 0.345 x cell (<0.0223 x +5V\_HD when on). Nominal R: on at 3.27-3.77 V, shed at 2.78-3.10 V 1 % R: on 3.23-3.82 V, shed 2.74-3.14 V; resistor hysteresis 0.327-0.340 V >= 0.31 V cell rebound.  
GPIO27 low or undriven: EN <= VF 0.715 V + 66 uA x 0.24k = 0.87 V < VENF 1.08 V, also with +5V\_HD still up. GPIO27 high: D2 reverse >= 1.7 V: 1 uA hot leakage moves EN 11 mV.  
Filter: 11.4 kOhm x 100 nF = 1.14 ms (140 Hz): the 96 kHz ESC ripple (0.21 V p-p at EN) falls to < 0.3 mV, D2 + R11 discharge it below VENF in 0.25 ms; turn-on from the clamp -1.6 mV at 4.2 V. U5B only (back-fed 2.5-2.8 V + 0.3 V ripple): EN <= 1.07 V < VENV. EN <= 1.62 V < 1.8 V abs max.

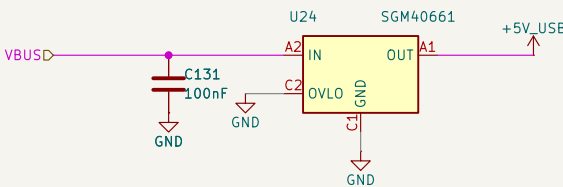
## CAMERA 5 V FILTER



Source: OpenFC-Lite-Mini/hardware/power.kicad\_sch (camera 5 V feed), with the commons 120 Ohm 0201 bead (blocks/LDO\_IMU\_TPS7A203S).

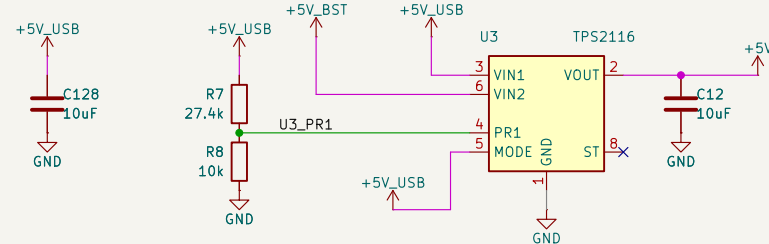
FB1 BLM03PX1225N1D: 120 Ohm at 100 MHz, 0.9 A, 160 mOhm: the 0.12 A camera drops 19 mV; it keeps the boost's 1 MHz and the camera's own switching apart. C15 10 uF X6S after the bead. +5V\_CAM feeds the CAM pad J24 and the camera plug J32 (PAD5); PWR\_FLAG because FB1 is passive. Feedback BLM03PX2205N1D (1.45 A).

## USB INLET OVP SGM40661



New block from the SGMICRO SGM40659/60/61 data sheet REV.A (D57).  
 IN = VBUS (J31 pin 4, PADS). OUT = +5V\_USB. OVLO (C2) to GND selects the internal threshold:  
 5.75–6.12 V rising, 5.45–5.66 V falling, IN 28 V abs max, 2.7–20 V recommended (feature limit: 40 V surge immunity), OVULO 2.4 V, 15 ms debounce, then a 2 ms soft turn-on: a cable LC ring (simulated <= 10.8 V at IN) never reaches TPS2116 VIN1 (6 V abs max), and the soft turn-on is the VBUS surge limit USB asks for above 10 V downstream.  
 C131 100 nF X65 25 V at the IN balls (data sheet CIN 0.1 uF; also takes ESD charge). COUT = C132 10 uF at U3 VIN1 (<= 100 uF allowed). RON 27 / 45 mOhm; 11–19 mV at 0.42 A. TA = 85 C.  
 Open: a non-compliant source at 5.5–6.12 V would pass before the OVLO trips (TPS2116 VIN1 5.5 V recommended, 6 V abs max); USB sources are <= 5.25 V (5.5 V with PD).

## 5 V POWER MUX TPS2116



New block from TI SLVSFG1A, priority mode (D24). C128 = the VIN1 cap: place it at U3 pin 3 (TI 9-10).  
MODE tied to VIN1 (+5VUSB), PR1 = VIN1 x 10k / 374k; USB is selected above VREF x 3.74 = 3.44-4.04 V (VREF 0.92-1.08 V;  
3.37-4.10 V with 1 % R), +5VBST (VIN2) otherwise. ST (open drain) unused.  
P4-2 = +5VBST holds only U2 VOUT/MODE, C8-C11, R4, U3 VIN2 and U5 VIN (netlist checked). USB selected: VIN2 stays off (truth  
table 'VIN2 = X') and both channels block reverse current, so USB never feeds +5VBST or the O4.  
P4-3: C128 10 uF x6S 0402 at VIN1 (TI CIN 1 uF); VIN1/MODE/PR1 peaks are bounded by U24. Unplug with a cell: +5V follows  
VB1 to its PR1 threshold, then the 8 us break-before-make gap at 0.7 V, then 8 us at C12 + C13 + C15, 3.4 uF each at 3.4  
V, then the curve to +5V = 3.4 V typical; worst corner (3.44 V - 2.75 V) = 0.69 V. C12 + C13 + C15 >= 3.07 V: the  
LP5912 blocks reverse current and its >= 7 uF output carries 0.2 A for 8 us, V1 case (b) scopes it.  
C12 10 uF at VOUT = +5V hold-up: Mux 40/60 mOhm: <= 0.04 V at 0.70 A analog; HD about 0.25 A (the O4 is on +5VBST ahead  
of the mux, D53 BR+10).

## SHEET NOTES

Power sheet, generated with hardware/tools/sch\_build.py + sch\_root.py (P4 power sheet); do not edit by hand.  
R15 are power symbols (net = Value): +BATT\_IN, +BATT, +5V\_BST, +5V\_USB, +5V, +5V\_CAM, +5V\_HD, +3V3, GND. Sheet pins: HD\_EN (in), CURR\_SENSE (out), VBUS (in). Local nets: SHUNT\_SENSE\_P(N/RC), U2\_SW\_U2\_FB, U3\_PR1\_U3\_EN.  
Closed here: P4-1, P4-2, P4-3, P4-7, P4-9, P4-12 (+3V3), P4-17 (BR-14 C13), P4-19 (C30 deleted); R11 is the GPIO27 pull-down P4-4 counts on; P4-13 is not used (D69; C2, C5, C14 stay X5R, sized on DC bias). Values: research/HOT-LOOP-CMP.md 4.1, 4.2, 15.5; P4 hand calculations  
P4 critique round 1 (D77): C30 (ESC3 EFMB VDD bulk) deleted; each ESC now has one 10 uF hot-design spec per AGM210MAP on its own sheet, and ESC3's slot-M cap C35 sits within 4 mm of UB VDD pin 4 (BR-05 as amended, 100 nF C31 at the pin). C17 (US C11) deleted: C8-C11 serve, P5 placement condition above.

# OpenDrone

Sheet: /POWER/  
File: power.kicad.

Title: Power: input, current sense, 5 V, 3.3 V, 04

Size: A3

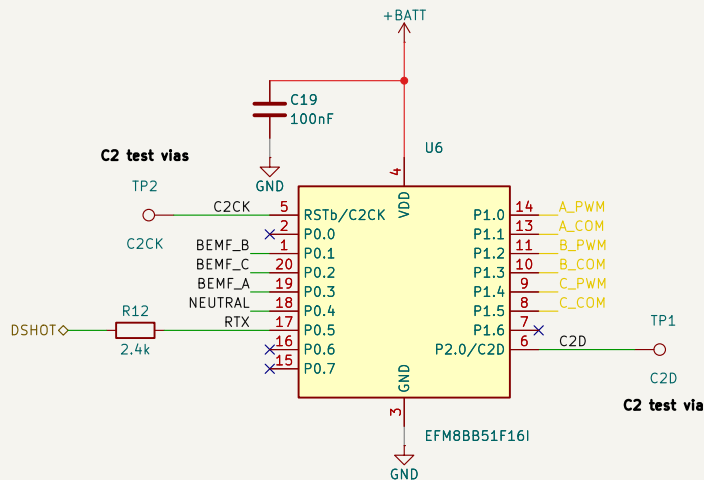
Date: 2026-10-09

Rev: rev1

KiCad E.D.A. 10.0.6

id: 2/14

## ESC MCU: EFM8BB51, Bluejay BB51 layout A



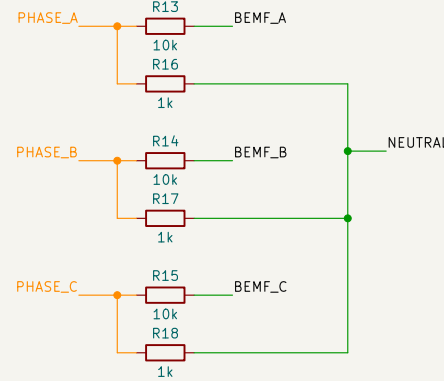
Pin use = Bluejay src/Layouts/BB51/A.inc (0368d11): P0.1 B\_Mux, P0.2 C\_Mux, P0.3 A\_Mux, P0.4 V\_Mux (neutral), P0.5 RTX; P1.0/P1.1 A\_Pwm/A\_Com, P1.2/P1.3 B, P1.4/P1.5 C (PWM active high = N gate, COM active low = P gate, all push-pull); P2.0 DebugPin = C2D; Pin numbers: EFM8BB51 data sheet Rev 1.0 Table 6.1. P0.0, P0.6, P0.7, P1.6 unused: no-connect (PINMAP 7.1).

VDD pin 4 straight on +BATT, no series R (it must track the P sources); C19 100 nF X6S at pin 4 / GND pin 3. The data sheet's 1 uF (Fig. 5.1) is the phase B hot-loop cap C23 10 uF X6S (about 3-4 uF effective at 4.2 V); BR-05 as amended (P4 critique): within 4 mm of pin 4 through the +BATT plane, 100 nF at the pin (v3 cell site 3.4-3.9 mm). ESC3 has no extra cap (C30 deleted, D77). VDD abs max 5.5 V vs 4.35 V cell + ripple + regen 4.7-5.25 V (V3c).

DShot (P4-B): R12 2.4k at RTX. At a 2.5 V cell the 3.3 V DShot high sits up to 0.53 V over P0.5's VDD + 0.3 V abs max; R12 holds the clamp current to  $\leq 0.35 \text{ mA}$   $((3.33 - 2.5) \text{ V} / 2.4\text{k})$ ; about 0.1-0.2 mA after the clamp drop) against 50 mA per pin. RSTb/C2CK: internal pull-up only (PINMAP F12).

C2D / C2CK: TP1 / TP2, mask-open filled test vias (D67), bench recovery only; NextPCB pre-programs Bluejay before reflow, later updates go through Betaflight passthrough on DSHOT.

## BEMF network: 10k series + 1k neutral star



Source: tinyPEPPER single\_esc.sch (CERN-OHL v1.2, wire trace in research 08): 10k series from each phase to its comparator input (BEMF\_A P0.3, BEMF\_B P0.1, BEMF\_C P0.2) and 1k from each phase to the virtual neutral (P0.4 V\_Mux, CMP0 negative). No leg to GND: at 1S the phase stays within the cell + one body-diode drop, and the 10k limit the input clamp current to  $< 0.1 \text{ mA}$ , 1k star, worst case one phase low and two high:  $2/3 \times 5.25 \text{ V} / 1\text{k} = 3.5 \text{ mA}$ , 12 mW per 0201 (50 mW part). P4-B: the netlist puts BEMF\_A/B/C and NEUTRAL on U6 pins 19/1/20 and 18 = A.inc A\_Mux 3 / B\_Mux 1 / C\_Mux 2 / V\_Mux 4.

ESC channel, one file placed 4x. The references drawn are ESC1's; the root sheet offsets them per instance (ESC2 U+1 Q+3 R+7 C+6 TP+2, ESC3 x2, ESC4 x3), matching bom\_plan.json:

ESC1 M1 rear right: U6 C19, hot loop C20 C23 C21, R12-R18, Q29-Q31, TP1/TP2

ESC2 M2 front right: U7 C25, hot loop C26 C29 C27, R19-R25, Q32-Q34, TP3/TP4

ESC3 M3 rear left: U8 C31, hot loop C32 C35 C33, R26-R32, Q35-Q37, TP5/TP6

ESC4 M4 front left: U9 C37, hot loop C38 C41 C39, R33-R39, Q38-Q40, TP7/TP8

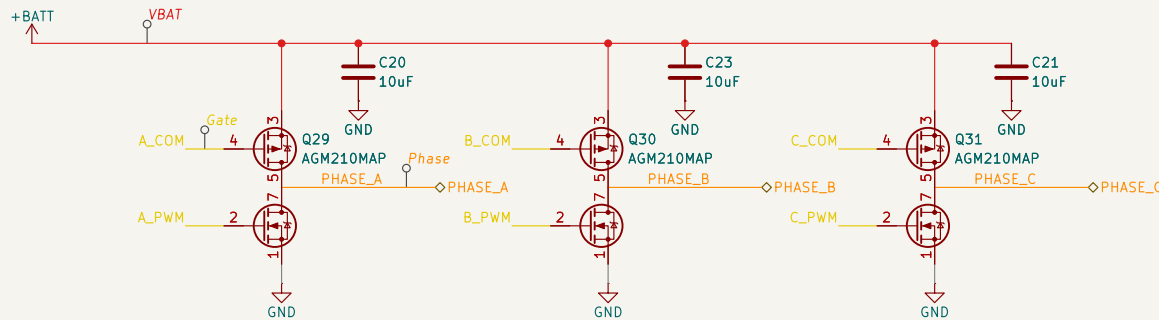
Pins: DSHOT (left, root net /MOTORn: FC GPIO25/24/23/22); PHASE\_A/B/C (right, local label + hierarchical label, net /ESCn/PHASE\_x, motor lands J4/J7/J10/J13). In-sheet nets: x\_COM, x\_PWM (Gate class), BEMF\_x, NEUTRAL, RTX, C2D, C2CK. Netclass flags: Phase (PHASE\_A), Gate (A\_COM), VBAT (+BATT rail): setup\_board.py patterns /ESC?/PHASE\_?, /ESC?/?\_COM, /ESC?/?\_PWM assign the rest.

Source: copied from OpenAIO hardware/esc\_channel.kicad\_sch (frames, block titles, C2 test points at the MCU pins, three-column BEMF block, power stage along the bottom) and adapted: AT32F421 + NSG2065Q + 6x DOY180N03T become EFM8BB51 + 3x AGM210MAP (tinyPEPPER direct drive, Matrix 1S 5IN1 II stage, D56), the 3S divider becomes the tinyPEPPER 1S star.

Power-up (P4-11, PINMAP F16): battery first, the EFM8 runs from the cell within 0.4 ms (POR 1.76 V, tPOR  $\leq 390 \text{ us}$ ) and the Bluejay bootloader turns on the 70-220k weak pull-ups (RTX included) while the RP2354A IOVDD still ramps (boost soft start + LDO, about 1-2 ms). Worst-case FC pad voltage in that window: the cell, up to 4.35 V (LiHV full), through  $\geq 72\text{k}$  (pull-up + R12), i.e.  $\leq 60 \text{ uA}$ ; 0.72 V over the 3.63 V unpowered-FT limit, current-limited. Accepted on Matrix parity (G473 FT pads about 4 V unpowered); V9 scopes the pad at power-up; fallback a 3.3 V-domain clamp or a DShot-path enable (not in rev1).

USB only: the PIO DShot idle-high back-feeds +BATT through the EFM8 I/O diode,  $\leq (3.3 \text{ V} - \text{VF}) / 2.4\text{k}$ , about 1 mA per channel; the EFM8s may cycle near POR, +5V stays on USB via the TPS2116 (10k fallback if V9 shows brown-out cycling).

## Half-bridges A/B/C: AGM210MAP P+N, one package per phase



AGM210MAP (AGMSEMI, DS VER2.73 p1 top view: D23 rule, P3 symbol/footprint check): 1 S1 N source = GND, 2 G1 N gate = x\_PWM, 3 S2 P source = +BATT, 4 G2 P gate = x\_COM, 5/6 + EP D2 (P drain) and 7/8 + EP D1 (N drain) = PHASE\_x; the two drain pads are joined only by the phase island on the board (never split it). Gates straight from the EFM8: no gate resistors and no OR options (tinyPEPPER / Matrix II field record; bom\_plan v3 has none); the about 0.15 A gate-edge current for tens of ns is a transient above the EFM8's 50 mA per-pin abs max, accepted as on the Matrix II (spec 4.4). VGS  $\pm 12 \text{ V}$ , VDS 20 V abs max vs VDD  $\leq 5.25 \text{ V}$  regen peak. Hot loop (D77, P4 critique): one cap per AGM210MAP, C20 / C23 / C21 10 uF X6S 0402 from pin 3 S2 (+BATT) to pin 1 S1 (GND) at the cell's lead side, both pads via-in-pad, so the commutation loop closes at the pins (Matrix II: one cap per FET); the three are also the ESC +BATT bulk. Bluejay: first power-up A\_X\_10\_96 (about 200 ns; DS td(off) + tf = 58 ns N at RGEN 3.3 Ohm, 84 ns P at 1 Ohm, longer on EFM8 port drive), then A\_X\_5\_96 (Matrix II) after the V2 shoot-through check (PINMAP 7.3); the DS Qg of 23 / 33 nC disagrees 3-5x with Ciss 950 pF, so V2 measures the gate edges.

# OpenDrone

Sheet: /ESC1/

File: esc\_channel.kicad\_sch

Title: ESC channel: EFM8BB51 + 3x AGM210MAP

Size: A4

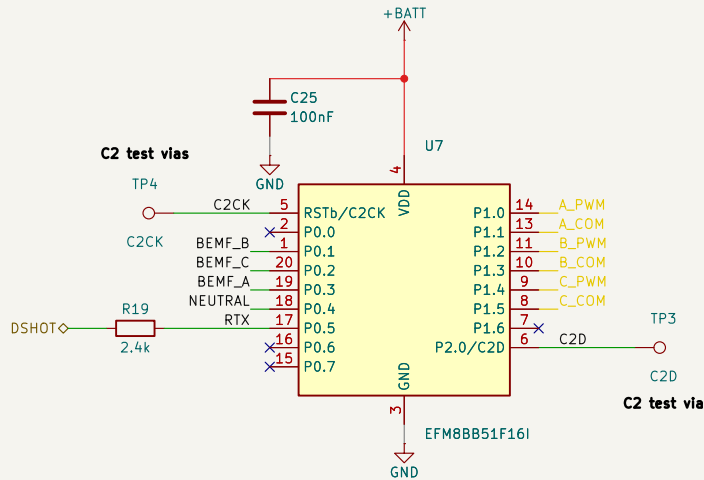
Date: 2026-10-09

Rev: rev1

KiCad E.D.A. 10.0.6

Id: 3/14

## ESC MCU: EFM8BB51, Bluejay BB51 layout A



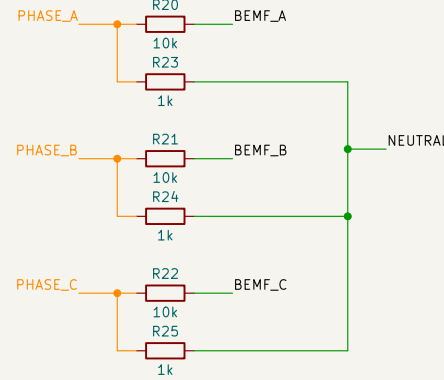
Pin use = Bluejay src/Layouts/BB51/A.inc (0368d11): P0.1 B\_Mux, P0.2 C\_Mux, P0.3 A\_Mux, P0.4 V\_Mux (neutral), P0.5 RTX; P1.0/P1.1 A\_Pwm/A\_Com, P1.2/P1.3 B, P1.4/P1.5 C (PWM active high = N gate, COM active low = P gate, all push-pull); P2.0 DebugPin = C2D. Pin numbers: EFM8BB51 data sheet Rev 1.0 Table 6.1. P0.0, P0.6, P0.7, P1.6 unused: no-connect (PINMAP 7.1).

VDD pin 4 straight on +BATT, no series R (it must track the P sources); C19 100 nF X6S at pin 4 / GND pin 3. The data sheet's 1 uF (Fig. 5.1) is the phase B hot-loop cap C23 10 uF X6S (about 3-4 uF effective at 4.2 V); BR-05 as amended (P4 critique): within 4 mm of pin 4 through the +BATT plane, 100 nF at the pin (v3 cell site 3.4-3.9 mm). ESC3 has no extra cap (C30 deleted, D77). VDD abs max 5.5 V vs 4.35 V cell + ripple + regen 4.7-5.25 V (V3c).

DShot (P4-B): R12 2.4k at RTX. At a 2.5 V cell the 3.3 V DShot high sits up to 0.53 V over P0.5's VDD + 0.3 V abs max; R12 holds the clamp current to  $\leq 0.35 \text{ mA}$   $((3.33 - 2.5) \text{ V} / 2.4\text{k})$ ; about 0.1-0.2 mA after the clamp drop) against 50 mA per pin. RSTb/C2CK: internal pull-up only (PINMAP F12).

C2D / C2CK: TP1 / TP2, mask-open filled test vias (D67), bench recovery only; NextPCB pre-programs Bluejay before reflow, later updates go through Betaflight passthrough on DSHOT.

## BEMF network: 10k series + 1k neutral star



Source: tinyPEPPER single\_esc.sch (CERN-OHL v1.2, wire trace in research 08): 10k series from each phase to its comparator input (BEMF\_A P0.3, BEMF\_B P0.1, BEMF\_C P0.2) and 1k from each phase to the virtual neutral (P0.4 V\_Mux, CMP0 negative). No leg to GND: at 1S the phase stays within the cell + one body-diode drop, and the 10k limit the input clamp current to  $< 0.1 \text{ mA}$ , 1k star, worst case one phase low and two high:  $2/3 \times 5.25 \text{ V} / 1\text{k} = 3.5 \text{ mA}$ , 12 mW per 0201 (50 mW part). P4-B: the netlist puts BEMF\_A/B/C and NEUTRAL on U6 pins 19/1/20 and 18 = A.inc A\_Mux 3 / B\_Mux 1 / C\_Mux 2 / V\_Mux 4.

ESC channel, one file placed 4x. The references drawn are ESC1's; the root sheet offsets them per instance (ESC2 U+1 Q+3 R+7 C+6 TP+2, ESC3 x2, ESC4 x3), matching bom\_plan.json:

ESC1 M1 rear right: U6 C19, hot loop C20 C23 C21, R12-R18, Q29-Q31, TP1/TP2

ESC2 M2 front right: U7 C25, hot loop C26 C29 C27, R19-R25, Q32-Q34, TP3/TP4

ESC3 M3 rear left: U8 C31, hot loop C32 C35 C33, R26-R32, Q35-Q37, TP5/TP6

ESC4 M4 front left: U9 C37, hot loop C38 C41 C39, R33-R39, Q38-Q40, TP7/TP8

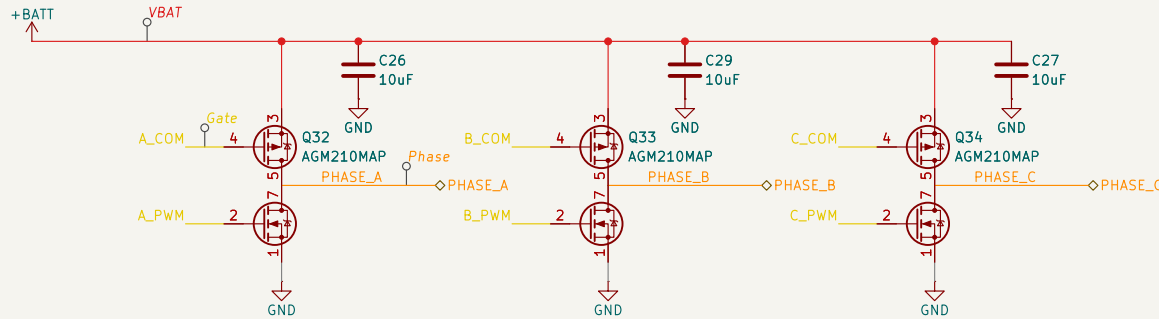
Pins: DSHOT (left, root net /MOTORn: FC GPIO25/24/23/22); PHASE\_A/B/C (right, local label + hierarchical label, net /ESCn/PHASE\_x, motor lands J4/J7/J10/J13). In-sheet nets: x\_COM, x\_PWM (Gate class), BEMF\_x, NEUTRAL, RTX, C2D, C2CK. Netclass flags: Phase (PHASE\_A), Gate (A\_COM), VBAT (+BATT rail): setup\_board.py patterns /ESC?/PHASE\_?, /ESC?/?\_COM, /ESC?/?\_PWM assign the rest.

Source: copied from OpenAIO hardware/esc\_channel.kicad\_sch (frames, block titles, C2 test points at the MCU pins, three-column BEMF block, power stage along the bottom) and adapted: AT32F421 + NSG2065Q + 6x DOY180N03T become EFM8BB51 + 3x AGM210MAP (tinyPEPPER direct drive, Matrix 1S 5IN1 II stage, D56), the 3S divider becomes the tinyPEPPER 1S star.

Power-up (P4-11, PINMAP F16): battery first, the EFM8 runs from the cell within 0.4 ms (POR 1.76 V, tPOR  $\leq 390 \text{ us}$ ) and the Bluejay bootloader turns on the 70-220k weak pull-ups (RTX included) while the RP2354A IOVDD still ramps (boost soft start + LDO, about 1-2 ms). Worst-case FC pad voltage in that window: the cell, up to 4.35 V (LiHV full), through  $\geq 72\text{k}$  (pull-up + R12), i.e.  $\leq 60 \text{ uA}$ ; 0.72 V over the 3.63 V unpowered-FT limit, current-limited. Accepted on Matrix parity (G473 FT pads about 4 V unpowered); V9 scopes the pad at power-up; fallback a 3.3 V-domain clamp or a DShot-path enable (not in rev1).

USB only: the PIO DShot idle-high back-feeds +BATT through the EFM8 I/O diode,  $\leq (3.3 \text{ V} - \text{VF}) / 2.4\text{k}$ , about 1 mA per channel; the EFM8s may cycle near POR, +5V stays on USB via the TPS2116 (10k fallback if V9 shows brown-out cycling).

## Half-bridges A/B/C: AGM210MAP P+N, one package per phase



AGM210MAP (AGMSEMI, DS VER2.73 p1 top view: D23 rule, P3 symbol/footprint check): 1 S1 N source = GND, 2 G1 N gate = x\_PWM, 3 S2 P source = +BATT, 4 G2 P gate = x\_COM, 5/6 + EP D2 (P drain) and 7/8 + EP D1 (N drain) = PHASE\_x; the two drain pads are joined only by the phase island on the board (never split it). Gates straight from the EFM8: no gate resistors and no OR options (tinyPEPPER / Matrix II field record; bom\_plan v3 has none); the about 0.15 A gate-edge current for tens of ns is a transient above the EFM8's 50 mA per-pin abs max, accepted as on the Matrix II (spec 4.4). VGS  $\pm 12 \text{ V}$ , VDS 20 V abs max vs VDD  $\leq 5.25 \text{ V}$  regen peak. Hot loop (D77, P4 critique): one cap per AGM210MAP, C20 / C23 / C21 10 uF X6S 0402 from pin 3 S2 (+BATT) to pin 1 S1 (GND) at the cell's lead side, both pads via-in-pad, so the commutation loop closes at the pins (Matrix II: one cap per FET); the three are also the ESC +BATT bulk. Bluejay: first power-up A\_X\_10\_96 (about 200 ns; DS td(off) + tf = 58 ns N at RGEN 3.3 Ohm, 84 ns P at 1 Ohm, longer on EFM8 port drive), then A\_X\_5\_96 (Matrix II) after the V2 shoot-through check (PINMAP 7.3); the DS Qg of 23 / 33 nC disagrees 3-5x with Ciss 950 pF, so V2 measures the gate edges.

# OpenDrone

Sheet: /ESC2/  
File: esc\_channel.kicad\_sch

Title: ESC channel: EFM8BB51 + 3x AGM210MAP

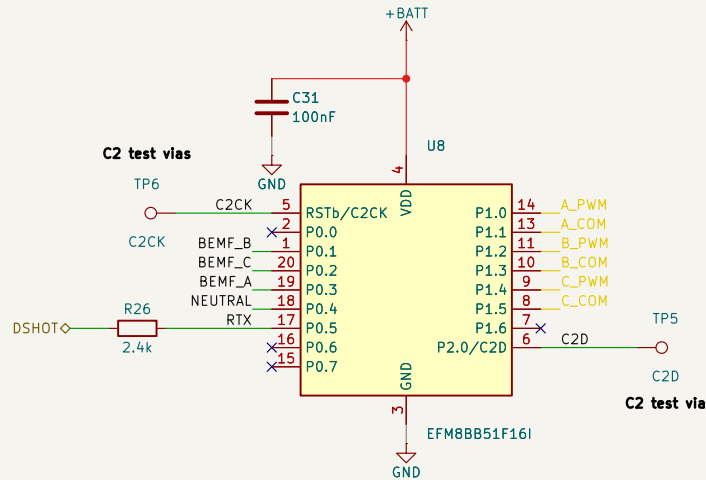
Size: A4 Date: 2026-10-09

KiCad E.D.A. 10.0.6

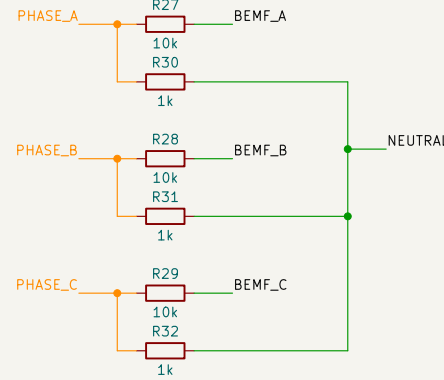
Rev: rev1

Id: 4/14

## ESC MCU: EFM8BB51, Bluejay BB51 layout A



## BEMF network: 10k series + 1k neutral star



Source: tinyPEPPER single\_esc.sch (CERN-OHL v1.2, wire trace in research 08): 10k series from each phase to its comparator input (BEMF\_A P0.3, BEMF\_B P0.1, BEMF\_C P0.2) and 1k from each phase to the virtual neutral (P0.4 V\_Mux, CMP0 negative). No leg to GND: at 1S the phase stays within the cell + one body-diode drop, and the 10k limit the input clamp current to  $< 0.1 \text{ mA}$ , 1k star, worst case one phase low and two high:  $2/3 \times 5.25 \text{ V} / 1\text{k} = 3.5 \text{ mA}$ , 12 mW per 0201 (50 mW part). P4-B: the netlist puts BEMF\_A/B/C and NEUTRAL on U6 pins 19/1/20 and 18 = A.inc A\_Mux 3 / B\_Mux 1 / C\_Mux 2 / V\_Mux 4.

ESC channel, one file placed 4x. The references drawn are ESC1's; the root sheet offsets them per instance (ESC2 U+1 Q+3 R+7 C+6 TP+2, ESC3 x2, ESC4 x3), matching bom\_plan.json:

ESC1 M1 rear right: U6 C19, hot loop C20 C23 C21, R12-R18, Q29-Q31, TP1/TP2

ESC2 M2 front right: U7 C25, hot loop C26 C29 C27, R19-R25, Q32-Q34, TP3/TP4

ESC3 M3 rear left: U8 C31, hot loop C32 C35 C33, R26-R32, Q35-Q37, TP5/TP6

ESC4 M4 front left: U9 C37, hot loop C38 C41 C39, R33-R39, Q38-Q40, TP7/TP8

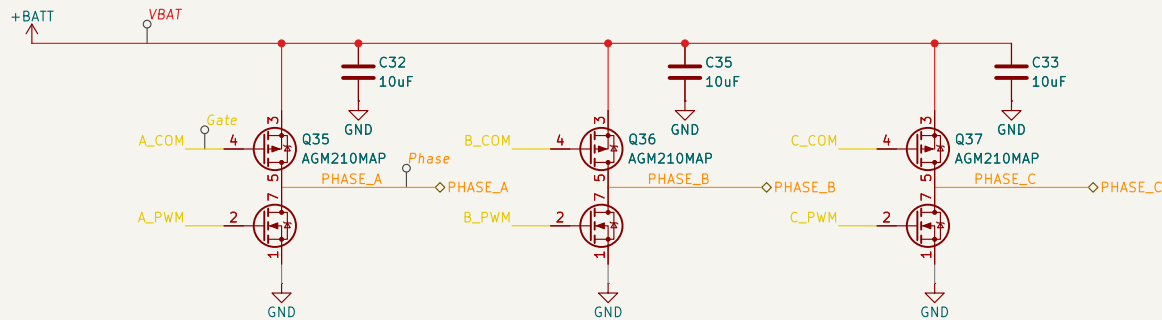
Pins: DSHOT (left, root net /MOTORn: FC GPIO25/24/23/22); PHASE\_A/B/C (right, local label + hierarchical label, net /ESCn/PHASE\_x, motor lands J4/J7/J10/J13). In-sheet nets: x\_COM, x\_PWM (Gate class), BEMF\_x, NEUTRAL, RTX, C2D, C2CK. Netclass flags: Phase (PHASE\_A), Gate (A\_COM), VBAT (+BATT rail): setup\_board.py patterns /ESC?/PHASE\_?, /ESC?/?\_COM, /ESC?/?\_PWM assign the rest.

Source: copied from OpenAIO hardware/esc\_channel.kicad\_sch (frames, block titles, C2 test points at the MCU pins, three-column BEMF block, power stage along the bottom) and adapted: AT32F421 + NSG2065Q + 6x DOY180N03T become EFM8BB51 + 3x AGM210MAP (tinyPEPPER direct drive, Matrix 1S 5IN1 II stage, D56), the 3S divider becomes the tinyPEPPER 1S star.

Power-up (P4-11, PINMAP F16): battery first, the EFM8 runs from the cell within 0.4 ms (POR 1.76 V, tPOR  $\leq 390 \text{ us}$ ) and the Bluejay bootloader turns on the 70-220k weak pull-ups (RTX included) while the RP2354A IOVDD still ramps (boost soft start + LDO, about 1-2 ms). Worst-case FC pad voltage in that window: the cell, up to 4.35 V (LiHV full), through  $\geq 72\text{k}$  (pull-up + R12), i.e.  $\leq 60 \text{ uA}$ ; 0.72 V over the 3.63 V unpowered-FT limit, current-limited. Accepted on Matrix parity (G473 FT pads about 4 V unpowered); V9 scopes the pad at power-up; fallback a 3.3 V-domain clamp or a DShot-path enable (not in rev1).

USB only: the PIO DShot idle-high back-feeds +BATT through the EFM8 I/O diode,  $\leq (3.3 \text{ V} - \text{VF}) / 2.4\text{k}$ , about 1 mA per channel; the EFM8s may cycle near POR, +5V stays on USB via the TPS2116 (10k fallback if V9 shows brown-out cycling).

## Half-bridges A/B/C: AGM210MAP P+N, one package per phase



AGM210MAP (AGMSEMI, DS VER2.73 p1 top view: D23 rule, P3 symbol/footprint check): 1 S1 N source = GND, 2 G1 N gate = x\_PWM, 3 S2 P source = +BATT, 4 G2 P gate = x\_COM, 5/6 + EP D2 (P drain) and 7/8 + EP D1 (N drain) = PHASE\_x; the two drain pads are joined only by the phase island on the board (never split it). Gates straight from the EFM8: no gate resistors and no OR options (tinyPEPPER / Matrix II field record; bom\_plan v3 has none); the about 0.15 A gate-edge current for tens of ns is a transient above the EFM8's 50 mA per-pin abs max, accepted as on the Matrix II (spec 4.4). VGS  $\pm 12 \text{ V}$ , VDS 20 V abs max vs VDD  $\leq 5.25 \text{ V}$  regen peak. Hot loop (D77, P4 critique): one cap per AGM210MAP, C20 / C23 / C21 10 uF X6S 0402 from pin 3 S2 (+BATT) to pin 1 S1 (GND) at the cell's lead side, both pads via-in-pad, so the commutation loop closes at the pins (Matrix II: one cap per FET); the three are also the ESC +BATT bulk. Bluejay: first power-up A\_X\_10\_96 (about 200 ns; DS td(off) + tf = 58 ns N at RGEN 3.3 Ohm, 84 ns P at 1 Ohm, longer on EFM8 port drive), then A\_X\_5\_96 (Matrix II) after the V2 shoot-through check (PINMAP 7.3); the DS Qg of 23 / 33 nC disagrees 3-5x with Ciss 950 pF, so V2 measures the gate edges.

# OpenDrone

Sheet: /ESC3/

File: esc\_channel.kicad\_sch

Title: ESC channel: EFM8BB51 + 3x AGM210MAP

Size: A4

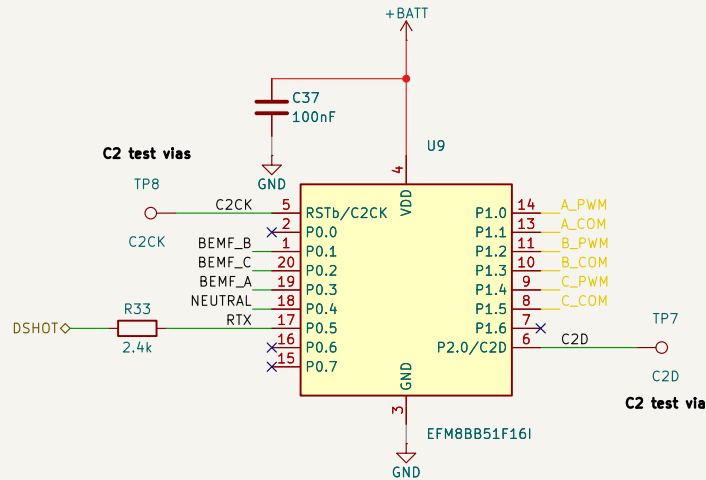
Date: 2026-10-09

Rev: rev1

KiCad E.D.A. 10.0.6

Id: 5/14

## ESC MCU: EFM8BB51, Bluejay BB51 layout A



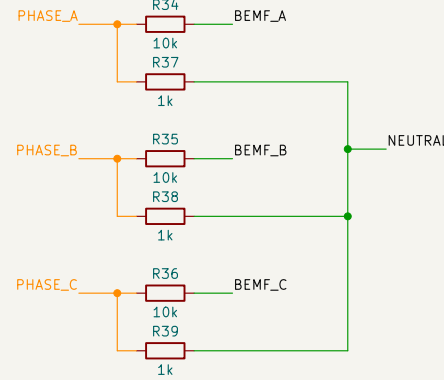
Pin use = Bluejay src/Layouts/BB51/A.inc (0368d11): P0.1 B\_Mux, P0.2 C\_Mux, P0.3 A\_Mux, P0.4 V\_Mux (neutral), P0.5 RTX; P1.0/P1.1 A\_Pwm/A\_Com, P1.2/P1.3 B, P1.4/P1.5 C (PWM active high = N gate, COM active low = P gate, all push-pull); P2.0 DebugPin = C2D. Pin numbers: EFM8BB51 data sheet Rev 1.0 Table 6.1. P0.0, P0.6, P0.7, P1.6 unused: no-connect (PINMAP 7.1).

VDD pin 4 straight on +BATT, no series R (it must track the P sources); C19 100 nF X6S at pin 4 / GND pin 3. The data sheet's 1 uF (Fig. 5.1) is the phase B hot-loop cap C23 10 uF X6S (about 3-4 uF effective at 4.2 V); BR-05 as amended (P4 critique): within 4 mm of pin 4 through the +BATT plane, 100 nF at the pin (v3 cell site 3.4-3.9 mm). ESC3 has no extra cap (C30 deleted, D77). VDD abs max 5.5 V vs 4.35 V cell + ripple + regen 4.7-5.25 V (V3c).

DShot (P4-B): R12 2.4k at RTX. At a 2.5 V cell the 3.3 V DShot high sits up to 0.53 V over P0.5's VDD + 0.3 V abs max; R12 holds the clamp current to  $\leq 0.35 \text{ mA}$   $((3.33 - 2.5) \text{ V} / 2.4\text{k})$ ; about 0.1-0.2 mA after the clamp drop) against 50 mA per pin. RSTb/C2CK: internal pull-up only (PINMAP 12).

C2D / C2CK: TP1 / TP2, mask-open filled test vias (D67), bench recovery only; NextPCB pre-programs Bluejay before reflow, later updates go through Betaflight passthrough on DSHOT.

## BEMF network: 10k series + 1k neutral star



Source: tinyPEPPER single\_esc.sch (CERN-OHL v1.2, wire trace in research 08): 10k series from each phase to its comparator input (BEMF\_A P0.3, BEMF\_B P0.1, BEMF\_C P0.2) and 1k from each phase to the virtual neutral (P0.4 V\_Mux, CMP0 negative). No leg to GND: at 1S the phase stays within the cell + one body-diode drop, and the 10k limit the input clamp current to  $< 0.1 \text{ mA}$ , 1k star, worst case one phase low and two high:  $2/3 \times 5.25 \text{ V} / 1\text{k} = 3.5 \text{ mA}$ , 12 mW per 0201 (50 mW part). P4-B: the netlist puts BEMF\_A/B/C and NEUTRAL on U6 pins 19/1/20 and 18 = A.inc A\_Mux 3 / B\_Mux 1 / C\_Mux 2 / V\_Mux 4.

ESC channel, one file placed 4x. The references drawn are ESC1's; the root sheet offsets them per instance (ESC2 U+1 Q+3 R+7 C+6 TP+2, ESC3 x2, ESC4 x3), matching bom\_plan.json:

ESC1 M1 rear right: U6 C19, hot loop C20 C23 C21, R12-R18, Q29-Q31, TP1/TP2

ESC2 M2 front right: U7 C25, hot loop C26 C29 C27, R19-R25, Q32-Q34, TP3/TP4

ESC3 M3 rear left: U8 C31, hot loop C32 C35 C33, R26-R32, Q35-Q37, TP5/TP6

ESC4 M4 front left: U9 C37, hot loop C38 C41 C39, R33-R39, Q38-Q40, TP7/TP8

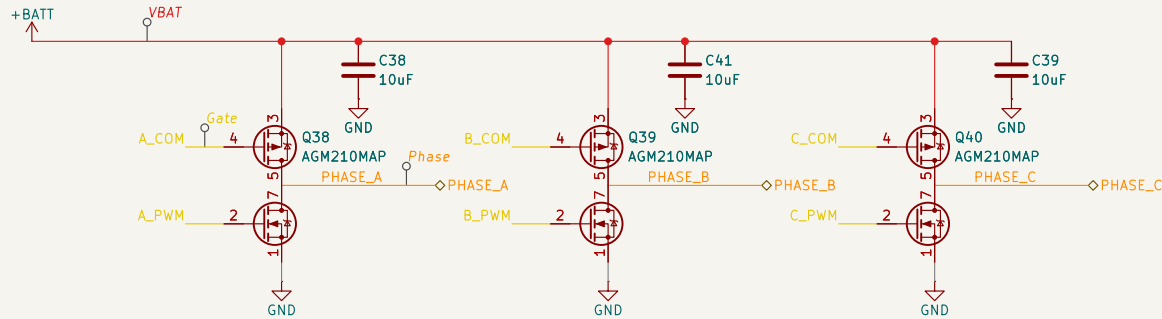
Pins: DSHOT (left, root net /MOTORn: FC GPIO25/24/23/22); PHASE\_A/B/C (right, local label + hierarchical label, net /ESCn/PHASE\_x, motor lands J4/J7/J10/J13). In-sheet nets: x\_COM, x\_PWM (Gate class), BEMF\_x, NEUTRAL, RTX, C2D, C2CK. Netclass flags: Phase (PHASE\_A), Gate (A\_COM), VBAT (+BATT rail): setup\_board.py patterns /ESC?/PHASE\_?, /ESC?/?\_COM, /ESC?/?\_PWM assign the rest.

Source: copied from OpenAIO hardware/esc\_channel.kicad\_sch (frames, block titles, C2 test points at the MCU pins, three-column BEMF block, power stage along the bottom) and adapted: AT32F421 + NSG2065Q + 6x DOY180N03T become EFM8BB51 + 3x AGM210MAP (tinyPEPPER direct drive, Matrix 1S 5IN1 II stage, D56), the 3S divider becomes the tinyPEPPER 1S star.

Power-up (P4-11, PINMAP F16): battery first, the EFM8 runs from the cell within 0.4 ms (POR 1.76 V, tPOR  $\leq 390 \text{ us}$ ) and the Bluejay bootloader turns on the 70-220k weak pull-ups (RTX included) while the RP2354A IOVDD still ramps (boost soft start + LDO, about 1-2 ms). Worst-case FC pad voltage in that window: the cell, up to 4.35 V (LiHV full), through  $\geq 72\text{k}$  (pull-up + R12), i.e.  $\leq 60 \text{ uA}$ ; 0.72 V over the 3.63 V unpowered-FT limit, current-limited. Accepted on Matrix parity (G473 FT pads about 4 V unpowered); V9 scopes the pad at power-up; fallback a 3.3 V-domain clamp or a DShot-path enable (not in rev1).

USB only: the PIO DShot idle-high back-feeds +BATT through the EFM8 I/O diode,  $\leq (3.3 \text{ V} - \text{VF}) / 2.4\text{k}$ , about 1 mA per channel; the EFM8s may cycle near POR, +5V stays on USB via the TPS2116 (10k fallback if V9 shows brown-out cycling).

## Half-bridges A/B/C: AGM210MAP P+N, one package per phase



AGM210MAP (AGMSEMI, DS VER2.73 p1 top view: D23 rule, P3 symbol/footprint check): 1 S1 N source = GND, 2 G1 N gate = x\_PWM, 3 S2 P source = +BATT, 4 G2 P gate = x\_COM, 5/6 + EP D2 (P drain) and 7/8 + EP D1 (N drain) = PHASE\_x; the two drain pads are joined only by the phase island on the board (never split it). Gates straight from the EFM8: no gate resistors and no OR options (tinyPEPPER / Matrix II field record; bom\_plan v3 has none); the about 0.15 A gate-edge current for tens of ns is a transient above the EFM8's 50 mA per-pin abs max, accepted as on the Matrix II (spec 4.4). VGS  $\pm 12 \text{ V}$ , VDS 20 V abs max vs VDD  $\leq 5.25 \text{ V}$  regen peak. Hot loop (D77, P4 critique): one cap per AGM210MAP, C20 / C23 / C21 10 uF X6S 0402 from pin 3 S2 (+BATT) to pin 1 S1 (GND) at the cell's lead side, both pads via-in-pad, so the commutation loop closes at the pins (Matrix II: one cap per FET); the three are also the ESC +BATT bulk. Bluejay: first power-up A\_X\_10\_96 (about 200 ns; DS td(off) + tf = 58 ns N at RGEN 3.3 Ohm, 84 ns P at 1 Ohm, longer on EFM8 port drive), then A\_X\_5\_96 (Matrix II) after the V2 shoot-through check (PINMAP 7.3); the DS Qg of 23 / 33 nC disagrees 3-5x with Ciss 950 pF, so V2 measures the gate edges.

# OpenDrone

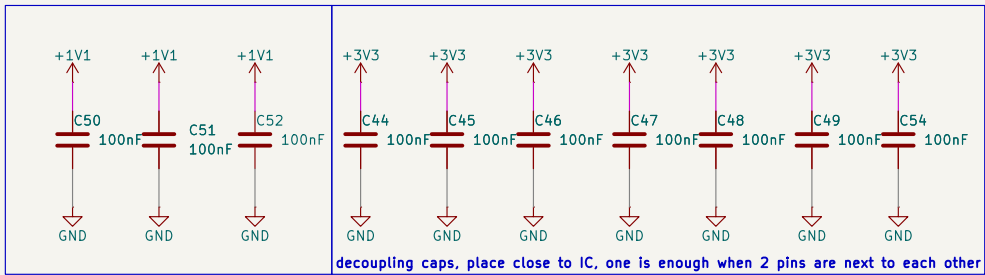
Sheet: /ESC4/  
File: esc\_channel.kicad\_sch

Title: ESC channel: EFM8BB51 + 3x AGM210MAP

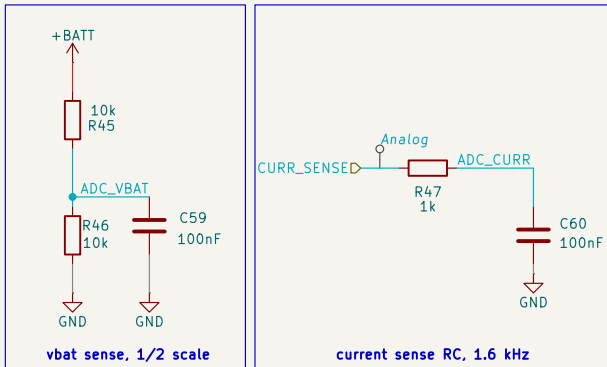
Size: A4 Date: 2026-10-09  
KiCad E.D.A. 10.0.6

Rev: rev1  
Id: 6/14

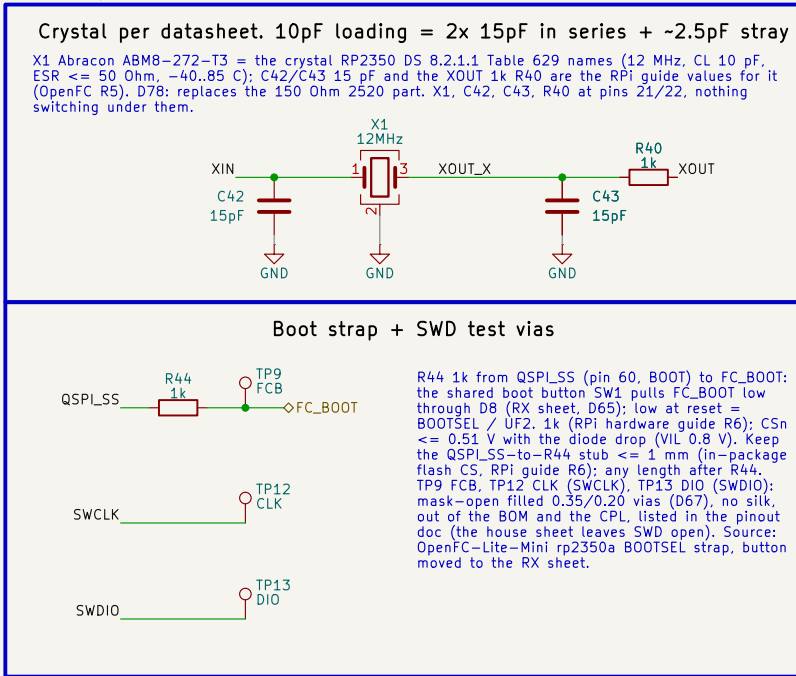
## RP2354A: Smaller variant with integrated flash (2MB)



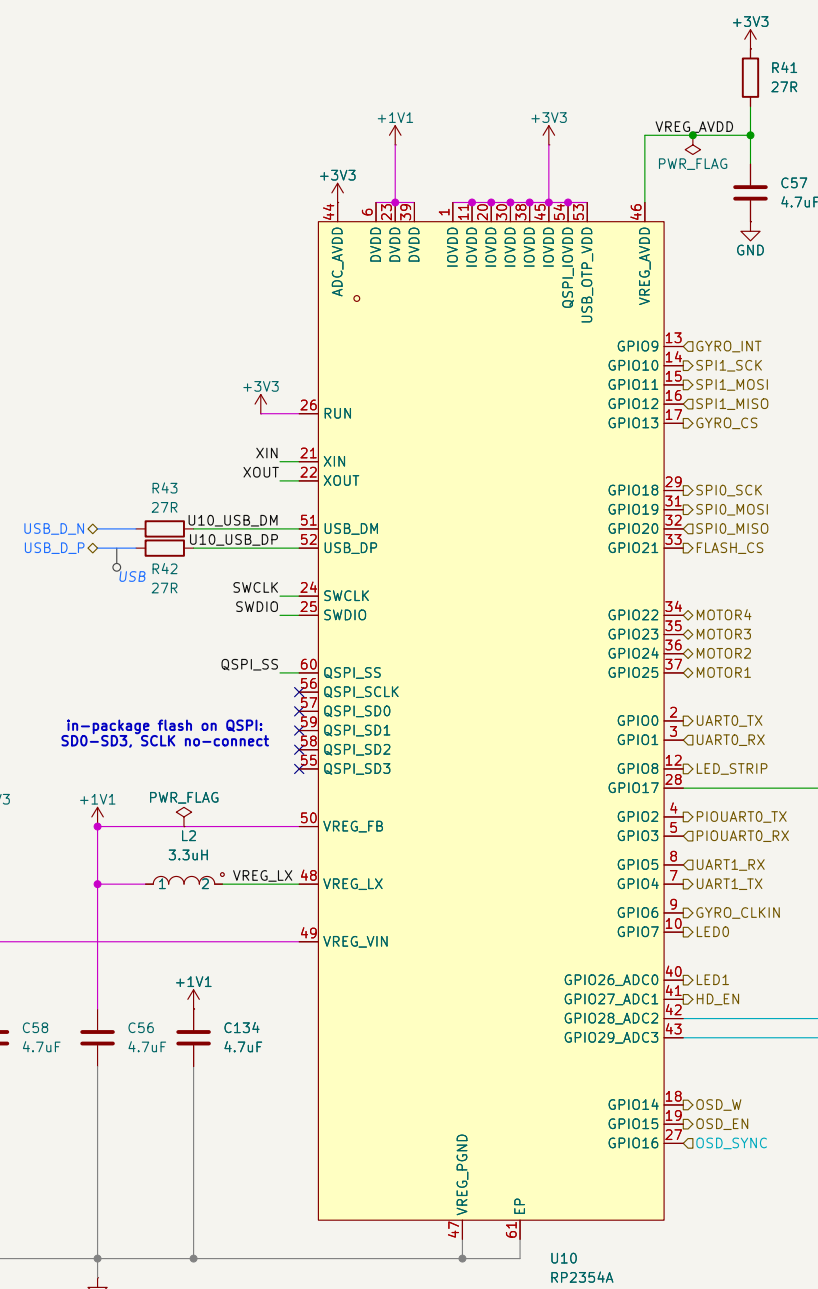
DVDD 6/23/39: C50–C52. IOVDD 1/11/20/30/38: C44–C48, one per pin. C49 sits at ADC\_AVDD pin 44 and also serves IOVDD 45 (BR–08: RP2350 DS 6.1.5: 100 nF close to ADC\_AVDD). C54 <= 1 mm from QSPL\_IOVDD pin 54 (the in-package flash supply, 2.97–3.63 V) and also serves USB\_OTP\_VDD 53 (BR–08, P4–17 placement condition). All Murata GRM033C91E104KE14D 100 nF X6S 0201 (BR–18). Copied from OpenFC–Lite–Mini rp2350a (7 caps for the 9 3V3 pins there too); its 4.7 uF on this row moved to the SMP5 block (C56).



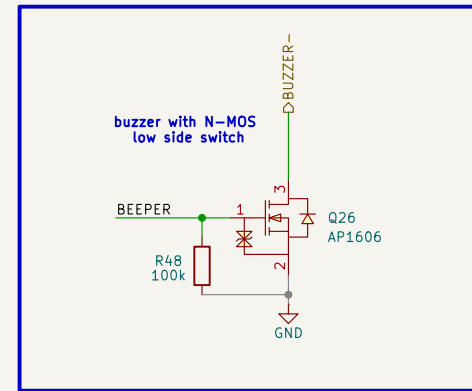
VBAT: R45/R46 10k/10k (house 100k/10k), 4.35 V => 2.18 V on GPIO29 (ADC3, not 5 V tolerant). vbat\_scale 200/100; 5 kOhm with C59 = 318 Hz, CURR: R47 1k + C60 100 nF = 1.6 kHz on GPIO28 (ADC2); INA186A3 on POWER; 50 mV/A, Ibatascale 500. Copied from OpenFC–Lite–Mini rp2350a (vbat + ESC current).



R42/R43 27R within 1–2 mm of pins 52/51 (RPI HW design 5.1, house 30R). USB\_D\_P/N: 90 ohm pair over unbroken GND, length-matched (D53); D7 ESD + J31 on PADS. Pin-side nets. U10\_USB\_DP/DM.



RP2350 1.1V DC–DC SMP5: follow the datasheet placement; RP2350 DS 6.3.8 (binding): C58 = CIN at VREG\_VIN, C56 = COUT on +1V1 with VREG\_FB taken from the C56 pad (not under L2); C58, L2 and C56 on the U10 side, smallest hot loops; CIN/COUT GND to VREG\_PGND at one point with 2 vias; no copper under L2 / VREG\_LX on the next layer. L2 A01A–B201610S3R3 is polarity-marked; orientation as DS Fig. 23/25 (pin 2 = VREG\_LX as on the flown house board). R41 27R + C57 4.7 uF filter VREG\_AVDD (1.25 kHz, BR–21; 3.135–3.63 V); C57 gets its own GND via to the EP. DS: CIN >= 4.7 uF, COUT 4.7 uF +/-20 %, C134 = the 2nd 4.7 uF on +1V1 that DS 6.3.8 recommends at DVDD pin 23 (fitted per the 2026–10–09 ruling, house board fits two); place it at pin 23.



Q26 AP1606 (copied from OpenFC–Lite–Mini rp2350a, Q5 there); 20 V, 0.7 A, Vgs(th) 0.4–1.2 V, RDS(on) <= 220 mOhm at Vgs 2.5 V (ALLPOWER DS p2). GPIO17 high = on (BEEPER\_INVERTED, active buzzer, no PWM); R48 100k holds the gate low through reset. BZ+ = +5V and BZ– = drain on the GND LED 5V BZ– pad row (PADS).

gyro SPI1 (IMU: GYRO\_CS pull–up R49 there)

blackbox SPI0 (FLASH\_CS pull–up R50 there)

PI00 bidir, DShot to the EFM8s (2.4k at the EFM8)

for CRSF receiver (ESP32 ELRS, UART0)

WS2812 pad LED, driven direct (F1, BR–04)

for external: user pads TP0/RP0 (PIO1, 8N1)

for MSP DisplayPort: DJI O4 Lite TX1/RX1

gyro CLKIN (ICM–42688–P only)

status LEDs (LED sheet, sinking)

HD line PINIO1: high = 04 on, analog VTX off

PIO2 framebuffer OSD: W, EN, SYNC consecutive

SHEET RP2350A, source: OpenFC–Lite–Mini/hardware/rp2350a.kicad\_sch (flown Rev2), copied block by block with its layout, frames and text style. Changes: USB–C + CC removed (USB = JST SH J31 + DT on PADS; 27R series R42/R43 stay at U10, D52 BR–21 value); I2CO pull–ups removed (GPIO4/5 = UART1); the inverting WS2812 stage is not copied (F1, BR–04; GPIO8 drives the LED pad); status LEDs moved to the LED sheet; BOOTSEL button replaced by R44 + FC\_BOOT (shared button on RX); SWD to test vias (D67); VBAT divider 10k/10k; 12 MHz crystal = ABM8–272–13 (D78); VREG\_AVDD 27R; two 4.7 uF on +1V1 (C56 at VREG\_FB, C134 at DVDD 23). GPIO map: research/PINMAP.md 2–3 (30 of 30 GPIOs, no move); Betaflight config PINMAP 5.

P4–11 power–up and 5 V–tolerant pads (PINMAP 4.5, F16; RP2350 DS Table 1433); an FT pad takes <= 3.63 V while IOVDD is 0–1.8 V (4.2 V at 2.5 V, 5.5 V at 3.3 V); the ADC pads GPIO26–29 take <= IOVDD + 0.5 V.

- MOTOR1–4 (GPIO25–22, F1), battery first: each EFM8 weak pull–up (70–220 kOhm to VDD = cell) reaches the pad through its 2.4k before +3V3 is up (boost soft start + LP5912 ramp, about 1–2 ms). Worst case 4.35 V open circuit = 0.72 V over 3.63 V, from >= 72 kOhm (<= 60 uA). Matrix parity (G473 FT pins –4 V at VDD 0). V9 scopes it: fallback (F16): 3.3 V–domain clamp or DShot–path enable.
- ADC\_VBAT (GPIO29): R45/R46 put up to 2.18 V on the pad from 5 kOhm while IOVDD < 1.7 V, so the pad clamp to IOVDD conducts <= 0.3 mA during the ramp; House + Matrix parity; V9 scope.
- HD\_EN (GPIO27): R11 2.4k + the BAS16LD from O4\_EN (<= 90 uA) hold it at <= 0.22 V before and during reset; inside IOVDD + 0.5 V and below the AP1606 HD–gate Vgs(th) min 0.4 V (P4–4); driven high it is >= 2.62 V (VOH, 1.4 mA into R11) over Vgs(th) max 1.2 V.
- UART0, SPI0/1, gyro, OSD, CURR\_SENSE: every driver is on +3V3 = IOVDD, no excursion. UART1 TX1/RX1: the O4 is fed only through US (needs HD\_EN high), none, USB D+/D–: the host drives only after the device pull–up.
- PIOUART0 pads TP0/RP0, and TX1/RX1, used as the second hardware UART in analog builds (IO–SET row 8); a peripheral on the 5V pads is up before IOVDD: 3.3 V logic only (README); a 5 V TX on RP0 or RX1 (GPIO5) would exceed 3.63 V at power–up.

# OpenDrone

Sheet: /RP2350A/  
File: rp2350a.kicad\_sch

**Title: RP2354A flight controller MCU**

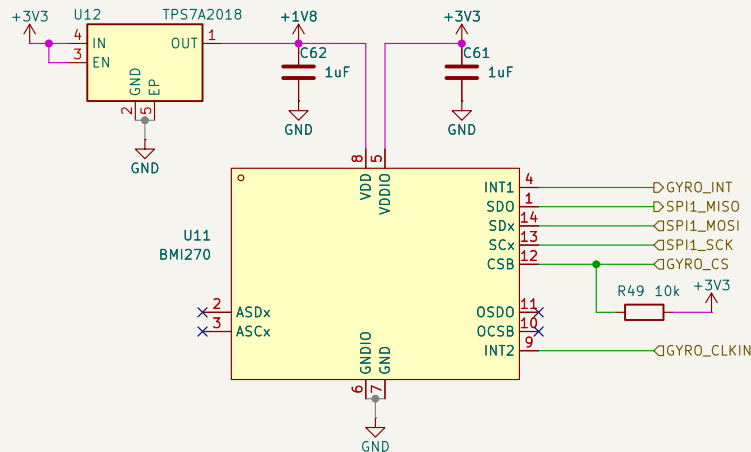
Size: A2 Date: 2026–10–09

Rev: rev1

KiCad E.D.A. 10.0.6

Id: 7/14

## Universal IMU wiring and LGA-14 land (BMI270, ICM-42688-P)



GYRO BMI270: U11, R49

Source: OpenFC-Lite-Mini imu.kicad\_sch (universal LGA-14 land, SPI1 4-wire, R49), copied and adapted. BMI270 on the BOM; TDK ICM-42688-P qualified on the same land (V10 fly-off, one part per revision).

Pins 2/3 (BMI270 ASDx/ASCx, TDK RESV) and 10/11 (OCSB/OSDO, TDK RESV): no-connect. BMI270 DS Table 22: unused ASDx/ASCx go to VDDIO or stay open, never to GND (the house sheet grounded them for the LSM6DSV16X).

Pins 6/7 GND (TDK pin 7 RESV must be GND). VDDIO pin 5 = +3V3 (RP2354A I/O level). VDD pin 8 = +1V8; the BMI270 has no supply-order rule (DS 4.3). Abs max 4.0 V on VDD/VDDIO, I/O up to VDDIO + 0.3 V.

Pin 9: TDK CLKIN = GYRO\_CLKIN (RP2354A GPIO6, PWM 32 kHz, driven only with the TDK part); BMI270 INT2, unused, left disabled. R49 10k; GPIO13 resets with its pull-down on, which would select the chip; R49 holds CSB >= 2.75 V.

P5: top side next to U12, LGA rotated 90 deg; GYRO\_1\_ALIGN from the placed orientation (PINMAP F11).

+1V8 GYRO LDO: U12, C61, C62 (D52 BR-09, P4-17)

Source: pcb-agent-commons blocks/LDO\_IMU\_TPS7A2033 (OpenFC-H7) topology with the 1.8 V part and no ferrite (fed from the +3V3 LDO): the house OpenFC-Lite-Mini also runs the gyro VDD from its own 1.8 V LDO. TPS7A2018 DQN: 1 OUT, 2 GND, 3 EN, 4 IN, 5 EP to GND. IN = EN = +3V3; VIN 1.6-6.0 V, VEN 0-6.0 V (abs max 6.5 V), EN never above IN. Load < 1 mA.

No 100 nF at the gyro: C61 = U12 CIN at BMI270 VDDIO pin 5, C62 = U12 COUT at VDD pin 8 (COUT >= 0.47 uF effective, ESR <= 100 mOhm; about 0.6-0.85 uF at 1.8 V). C61 counts in the +3V3 COUT sum (P4-12, POWER).

P5: U12 on the gyro side with OUT facing pin 8; C61 between U12 pin 4 and gyro pin 5, C62 <= 1.5 mm from pin 8 (P5-06), else re-place U12.

# OpenDrone

Sheet: /IMU/

File: imu.kicad\_sch

**Title: Gyro BMI270 + 1.8 V LDO**

Size: A5

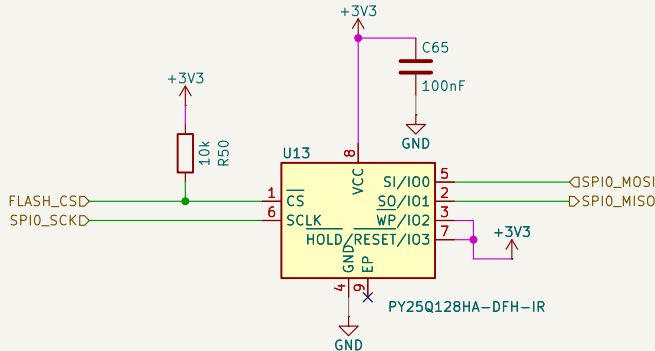
Date: 2026-10-09

Rev: rev1

KiCad E.D.A. 10.0.6

Id: 8/14

## SPI NOR PY25Q128HA: 16 MB blackbox on SPI0 (U13 C65 R50)



Placement (P5): beside the RP2354A on SPI0 (bom\_plan region nor, top; either side works). U13, SPI0 and FLASH\_CS stay  $\geq 3$  mm from the RX antenna hole (spec 4.9 and 11, DRU 'SPI0 3 mm from the RX antenna hole'): SPI0 harmonics fall in the ELRS band, V7 measures desense while logging. C65 at pin 8. Filled+capped 0.35/0.20 via-in-pad allowed in the lead lands, never in the tabs. Betaflight sets SCK to fast slew (bus\_spi\_pico.c); MOSI and CS keep the reset 4 mA drive and slow slew (EMI lever without parts). Sourcing: no LCSC listing; NextPCB buys by MPN (D60), the quote confirms it. Same-pinout fallbacks in the Betaflight table (USON-8 4x4 land not staged): PY25Q128HA-QVH-IR, GD25Q128EQIGR.

Source: OpenFC-Lite-Mini hardware/blackbox.kicad\_sch (rev2). Kept: layout, the 10k CS pull-up (R39  $\rightarrow$  R50) and the 100n supply cap (C43  $\rightarrow$  C65). Changed: its microSD (Card1) became this 16 MB SPI NOR (D66); the SPI0(SCK,MOSI,MISO) bus label became four hierarchical labels (sch\_contract.json).

U13 Puya PY25Q128HA-DFH-IR: 128 Mbit, USON-8 3x3 (DF), 2.7-3.6 V, -40..85 C, fC 133 MHz, READ 03h 80 MHz, JEDEC ID 85 20 18 is in the Betaflight m25p16 table: USE\_FLASH\_M25P16, SPI0 PA18/PA19/PA20, FLASH\_CS\_PIN PA21 (PINMAP). SPI0 runs at 75 MHz for writes AND reads: Betaflight bus\_spi\_pico.c spiCalculateDivider() uses an even prescale  $\geq 2$  on clk\_peri 150 MHz, so the table's 133 / 80 MHz both give 150/2. Writes are inside fC; reads are not guaranteed: tCLQV 6-7 ns vs the 6.67 ns SCK half period plus pad and trace delay (the RP2350 DS gives no SPI receive timing). Gate F7 / V9: erase, write, read back and CRC a full log, status polling under load; fallback = a lower read clock for this JEDEC ID (firmware cap or table entry), upstreamed with the board config.

WP# (3) and HOLD#/RESET# (7) tied to +3V3: the DS gives an internal pull-up only to the SOP-16 dedicated RESET#, so pin 7 must not float. Delivered SR = 00h, CR = 00h (QE 0, HOLD# function, WPS 0: no block lock) and Betaflight sends 1-bit SPI commands only, so IO2/IO3 never drive against the rail (the ERC pin\_to\_pin warning on pins 3/7 against a +3V3 power output is this deliberate tie).

R50 10k: CS# must follow VCC at power-up (DS 5.5). RP2354A pads reset with the pull-down on (PDE 1, RPD 36-113 kOhm at 3.3 V): CS  $\geq 36/46$  VCC = 0.78 VCC > VIH 0.7 VCC, so U13 stays deselected through reset and boot. U13 and the RP2354A IOVDD share +3V3: no pin above VCC + 0.5 V.

C65 100 nF X65 at pin 8, GND return straight to pin 4 (ICC max 13 mA read, 12 mA erase; adds 0.1 uF nominal to the +3V3 COUT sum of P4-12). EP (9) = two tie-bar tabs with no electrical function in the DS: no-connect, no vias in the tabs.

# OpenDrone

Sheet: /BLACKBOX/

File: blackbox.kicad\_sch

**Title: Blackbox 16 MB SPI NOR PY25Q128HA**

Size: A5

Date: 2026-10-09

Rev: rev1

KiCad E.D.A. 10.0.6

Id: 9/14

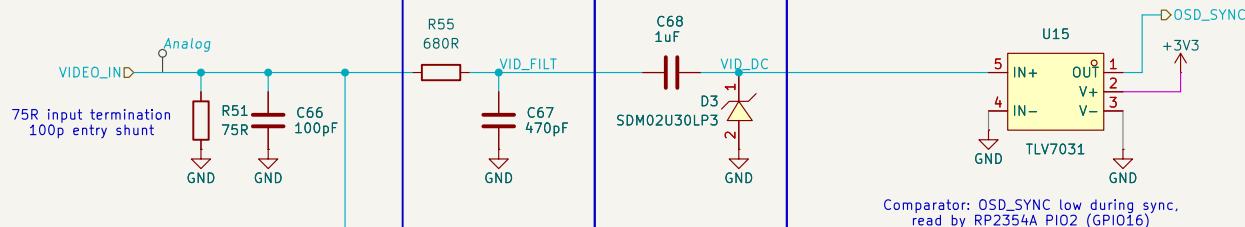
# VIDEO PATH

Chroma Filter  
fc = 500 kHz

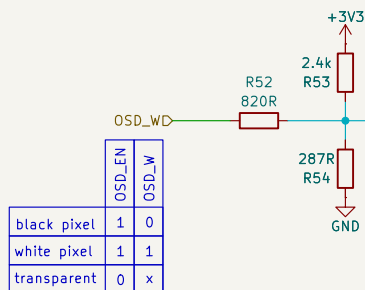
DC Restoration

Sync detection, VID\_DC sync tips  
clamped to  $-V_F$  (about  $-0.1...-0.2$  V)

U15 V+ (2) is decoupled by C70 in the switch cell,  
one 100n shared with U14 (BR-13)



Resistor divider lets the OSD\_W GPIO select  
white (high: 1.05 V) or black (low: 0.27 V)



S low (OSD\_EN low / reset): B1 camera  
S high: B2 OSD level

Analog switch

VIDEO\_OUT: the switch output drives the RTC6705 video network on the VTX sheet directly (R58 1.2k || C104 33p, R59 270R, C105 4.7u, R60 1.2k: OpenOSD-X values); the house COS8051 buffer (U1, R46-R48, C41) is deleted (spec 4.8). OSD levels (P4-10, recomputed at the P4 critique with the real load): open-circuit at OSD\_LVL (+3V3 3.30 V, OSD\_W push-pull) black 0.268 V / white 1.054 V from 195R (R53) | R54 = 0.352 V / 256R, with R52) + switch on 7-18R. VIDEO\_OUT is loaded by the VTX network ahead of C105: R58 + R59 = 1.47 kOhm at DC, about 0.95 kOhm at 5 MHz and 0.60 kOhm at 10 MHz (R58 / C104 4.0 MHz, 21.9 MHz with R59; 270R only above about 20 MHz). Loaded levels: black 0.234-0.236 V, white 0.920-0.927 V (swing x0.88 of the open-circuit 0.786 V); edge gain x0.86 at 5 MHz and x0.80 at 10 MHz vs x0.97 for the camera (37.5R source): OSD text slightly dimmer and softer than the camera picture, not a failure (the house COS8051 buffer that made the source low-impedance is deleted). V9 checks OSD white / black and edge sharpness on a monitor; fallback: rescale R52-R54 within the 3 mA OSD\_W budget. PS-01 (R54 270R) would lower white to about 0.89 V: not recommended. OSD\_W: 2.74 mA high ( $\leq 3$  mA, 2.79 mA at 3.36 V), 0.33 mA low.

Source: OpenFC-Lite-Mini hardware/osd.kicad\_sch (rev2, flown), copied and adapted: cells, texts, truth table and part spots kept (A3 -> A4). Same topology: 75R termination, 680R/470p chroma filter, 1u + SDM02U30LP3 sync-tip clamp, TLV7031 comparator, SN74LVC1G3157 camera/level switch (contract source, spec 4.8).

Changed: COS8051 buffer deleted (output = VIDEO\_OUT); divider R42 3.9k / R43 12k / R44 1.4k -> R52 820R / R53 2.4k / R54 287R (about 5x down, P4-10); C39 + C42 -> one shared C70 100n X65 (D52 BR-13 + BR-18); C66 100p entry shunt added (spec 11); refs per bom\_plan (R41>R51, C38>C67, C40>C68, D9>D3, U2>U15, R45>R55, R42-44>R52-54, U18>U14). Symbols from the OpenDrone catalogue with TI pin numbers: TLV7031 X2SON 1 OUT, 2 V+, 3 V-, (centre pad), 4 IN-, 5 IN+, SN74LVC1G3157 DTB 1 B2, 2 GND, 3 B1, 4 A, 5 VCC, 6 S. The house lib symbols used the EasyEDA pad order (consistent with its own footprints), so the drawing is re-routed with the same nets: IN+ = VID\_DC, IN- = GND, B1 = camera, B2 = OSD level. The house sheet's rotated ground is upright here.

Operation: OSD\_EN low (RP2354A reset pull-down, OSD idle) -> B1, the camera passes through. OSD\_EN high -> B2: OSD\_W high = white, low = black. No OSD without camera sync (osd\_pico.c overlays on detected sync). OSD\_W/EN/SYNC = GPIO14/15/16, consecutive, PIO2 (PINMAP).

Sync threshold (P4-10): 680R/470p = 498 kHz: colour burst x0.11-0.14 (+-17-21 mV), 4.7 us sync kept (tau 0.32 us). C68 + D3 (cathode on VID\_DC) clamp the sync tips to  $-V_F$ ; the leakage-limited clamp current (uA average, tens of uA at the tip) gives  $V_F$  about 0.1-0.2 V over temperature (I: DS 0.37 V max at 10 mA / 25 C, 0.20 V at 10 mA / 125 C). IN- at GND puts the trip point  $V_F$  above the tip, 33-67 % of the 0.3 V sync,  $\geq 0.08$  V from tip and blanking after VIO +8 mV and VHYS 2-17 mV. OSD\_SYNC is low during sync (osd\_tx.pio waits for a low sync pin). IN+ at  $-V_F$  stays inside abs max ( $V_{EE} - 0.3$  V), at most 0.1 V under the ROC input floor ( $V_{EE} - 0.1$  V), no phase reversal (DS 6.4.1). tpd 3 us on both edges (equal shift, as on the house board).

Ratings: +3V3 only (U14 1.65-5.5 V, U15 1.6-6.5 V). S: VIH 0.7 VCC = 2.31 V, edges  $\leq 10$  ns/V (fast slew pads). AC-coupled camera sync tips (-0.3 to -0.4 V) are below the switch ROC (0.VCC) but inside abs max (-0.5 V, 50 mA; 37.5R source), as on the house board: V9 checks. Supply: U15 0.9 uA, U14  $\leq 10$  uA, divider 0.94-1.26 mA + OSD\_W 2.74 mA. C68 X5R carries  $\leq 0.6$  V DC: no DC-bias loss (P4-13 hot-zone rule void, D69).

Placement (P5; P4-17 sheet note): bottom, region osd. BR-13: C70 is the only supply cap of U14 (VCC 5) and U15 (V+ 2): place it between the two parts (2.5 mm apart). GND via-In-pad; the P5 check lists its distance to both pins. U15 centre pad (3, V-) to GND through a via (TI 7.4). R51 and C66 at the video entry on the CAM pad J23 / plug J32 line: camera return into L2 at the CAM GND pad: video on L1/L6 over a plane with GND both sides, never L3 (spec 11). R55, C67, C68, D3 next to U15 IN+. Video parts away from the boost and its inductor (preference, D69; v2 floorplan 3.3-8.6 mm: V5 video test). Netclass Analog: VIDEO\_IN (flag here), VIDEO\_OUT, VID\_FILT, VID\_DC, OSD\_LVL, OSD\_SYNC by the setup\_board.py patterns: OSD\_W, OSD\_EN Default. Sourcing: exact MPNs from bom\_plan: LCSC empty where bom\_plan has none (C66, R52, R54; open item A9-LCSC). PS-01 (R54 -> 270R RC0201FR-07270RL, shared with R59, -1 line): black 0.258 V, white 1.011 V, 187R, OSD\_W 2.79 mA, also passes P4-10; proposed in contract-issues, drawn = bom\_plan.

# OpenDrone

Sheet: /OSD/  
File: osd.kicad\_sch

**Title: PIO analog OSD front end**

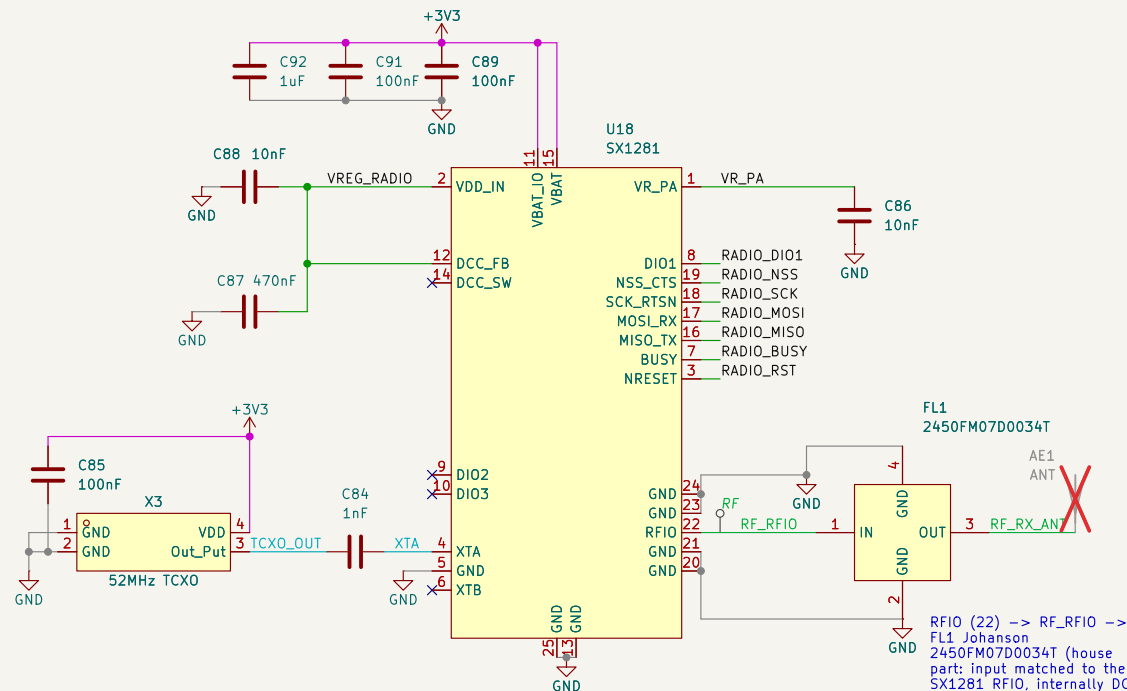
Size: A4 Date: 2026-10-09

KiCad E.D.A. 10.0.6

Rev: rev1

Id: 10/14

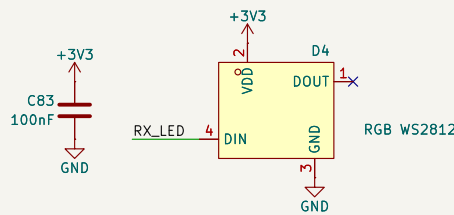
## SX1281 RADIO + 2.4 GHz FILTER + WIRE ANTENNA



SX1281 (U18), copied from OpenRX-Lite-UFL; SX1281IMLRT (D52 BR-25), SX1280IMLRT drop-in alternate (ranging engine only).  
LDO MODE (Semtech DS Rev 3.2 5.1, Fig. 15-4): DCC\_SW (14) open, the sibling's 15 uH L4 deleted (the ELRS layout has no radio\_dcdc); VDD\_IN (2) tied to DCC\_FB (12) = VREG\_RADIO with C87 470 nF + C88 10 nF; VR\_PA (1) C86 10 nF; VBAT\_IO (11) C91 and VBAT (15) C89 100 nF (Table 15-1) + C92 1 uF radio local (P4-12 lever PS-06); the house sheet's extra 10 nF / 10 uF deleted (BR-06).  
PIN 5 IS GND (DS pin table) and is grounded here; the sibling sheets left it open (only XTB stays open with a TCXO, DS 15.2). DIO2 / DIO3 open.  
52 MHz TCXO X3 (kept, D53): clipped sine AC-coupled by C84 1 nF into XTA (4). C85 100 nF at VCC, XTB (6) open.

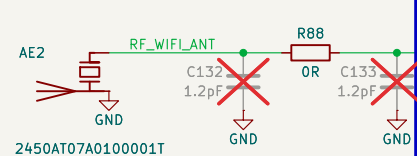
RFIO (22) -> RF\_RFIO -> FL1 Johanson 2450FM07D0034T (house part: input matched to the SX1281 RFIO, internally DC blocked, IN pin 1 toward the chip) -> RF\_RX\_ANT (50 Ohm) -> AE1: insulated lambda/4 wire (about 28-29 mm, trimmed in V7) in a plated D0.5 hole; AE1 DNP, out of BOM / CPL. RF class: board rule areas RF\_RX\_ANT / RF\_RX\_EXIT.

## ELRS RGB LED



D4 XINGLIGHT XL-1010RGB-2812B (copied from OpenRX-Lite-UFL) on +3V3, ELRS led\_rgb GPIO22, one LED (overlay ledidx\_rgb\_vtx [0]).  
P4-15 DECIDED: KEEP ON +3V3. Datasheet: VDD 3.5-5.5 V, DIN VIH >= 2.8 V, 5 mA per colour with a 0.6 V sink knee. On +3V3 the supply is 0.2 V under the table minimum (blue / green may run below 5 mA; Vf + knee is about 3.4-3.6 V); DIN comes from a 3.3 V CMOS output on the same rail. On +5V the supply would be in range but the data guarantee is short (ESP32 VOH >= 0.8 VDD = 2.64 V at a high-impedance load, DS Table 18, vs VIH 2.8 V). Neither option sits fully inside the tables; the house RX circuit (OpenRX-Lite, -UFL, OpenAIO) runs this LED on 3.3 V, so it stays. Bench V-LED on the first boards: red, green, blue and white at 3.3 V, cold and hot. Failback: VDD pin to +5V (one trace: C83 is 25 V).

## Wi-Fi CHIP ANTENNA (D58)

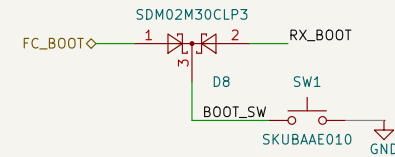


Wi-Fi (D58): LNA\_IN (2) -> RF\_WIFI 50 Ohm CPW <= 8 mm -> C133 shunt (DNP) / R88 OR series (fitted) / C132 shunt (DNP) -> AE2 Johanson 2450AT07A0100001T (FEED 1/4, GND 2/3). Same pi as Espressif Fig. 11 (C4/L1/C3) and OpenRX-Lite-UFL (C22/L2/C21). The PICO matches LNA\_IN to 50 Ohm inside, so the network only tunes the antenna: values from V7 tuning (VNA at the feed or Johanson's service); R88 may become a Murata LQP03TN 0201 inductor. The AE2 footprint carries the 5.0 x 3.2 mm keep-out (DRU 'D21 wifi\_antenna\_edge\_strip').

## NOTES

RX SHEET (P4 shard C). Sources: the SX1281 radio, 52 MHz TCXO, RGB LED and Wi-Fi pi match are COPIED from the house sheet OpenRX-Lite-UFL hardware/esp32c3\_sx1281\_lite.kicad\_sch (same as OpenAIO rx\_esp32c3\_sx1281.kicad\_sch); same symbol positions and wires, adapted to bom\_plan refs/values/MPNs and to D52/D53/D58 (LDO mode, pin 5 to GND, U:FL replaced by the wire-antenna hole, 2450AT07A0100001T chip antenna, grounds turned down); the house Johanson 2450FM07D0034T front-end filter stays. The ESP32-PICO-V3 block is new, from the Espressif ESP32-PICO series datasheet v1.3 Fig. 11 (D52 BR-01, D55); the shared boot / bind button is new (D65, Matrix II style). Pin plan: research/PINMAP.md 6. Parts: hardware/bom\_plan.json. Radio and LED +3V3 come from the POWER sheet (U4 LP5912-3.3).

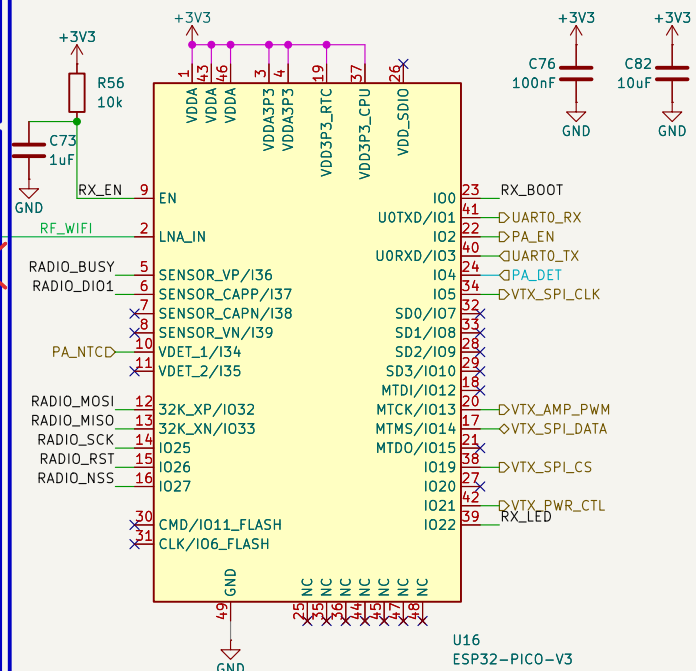
## BOOT / BIND BUTTON (D65)



BOOT / BIND BUTTON (D65: new). SW1 Alps SKUBAAE010 (2.4 x 1.4 mm) from BOOT\_SW to GND. D8 Diodes SDM02M30CLP3 dual Schottky, common cathode on BOOT\_SW; A1 = FC\_BOOT (RP2354A QSPILSS through R44 1k, RP2350A sheet), A2 = RX\_BOOT (ESP32 GPIO0, pin 23).  
Released: each strap sees only diode leakage (IR <= 0.15 uA at 10 V, 25 C), so GPIO0 keeps its internal pull-up and QSPILSS its own level.  
Held while power is applied: RP2354A BOOTSEL and ESP32 ROM download mode (CSn <= 0.51 V vs VIL 0.8 V; GPIO0 <= 0.42 V vs VIL 0.825 V); each mode stays latched until that chip's next reset.  
Production / ELRS recovery (D67, spec 12.1): (1) hold SW1 (or probe BOOT\_SW) while powering; (2) write the Betaflight UF2 (or picotool reboot): the RP2354A restarts into Betaflight, the ESP32 stays in download mode; (3) esptool through Betaflight serial passthrough.  
Pressed while running: ELRS button (bind 1.5 s, Wi-Fi 5 s, reboot 12 s); 2.9 mA through R44, CS stays >= 2.62 V, the FC keeps running.  
Bench V-btn: if GPIO0 dips below 2.5 V while released (QSPILSS edges under XIP load coupling through D8), add 100 pF 0201 on BOOT\_SW.

## ESP32-PICO-V3 MCU (ELRS)

ESP32-PICO-V3 (U16), Espressif PICO datasheet v1.3 Fig. 11 / Table 4 (new block).  
SUPPLY: VDDA (1, 43, 46), VDDA3P3 (3, 4), VDD3P3\_RTC (19), VDD3P3\_CPU (37) on +3V3; C82 10 uF + C76 100 nF at VDDA3P3 pins 3/4 (Fig. 11 C1/C2). The 4 MB flash, 40 MHz crystal, CAP1/CAP2 network and the VDD\_SDIO capacitor are inside the SIP (Fig. 8); VDD\_SDIO (26) stays open.  
EN: CHIP\_PU (9) = RX\_EN, R56 10k to +3V3 + C73 1 uF (tau 10 ms), never floating.  
STRAPS at reset (P4-6, checked on the netlist):  
GPIO0 (23) RX\_BOOT; internal 45k pull-up, D8 blocks -> 1 = SPI boot (SW1 held -> download).  
GPIO2 (22) PA\_EN; internal 45k pull-down + LP5907 EN 1 MOhm pull-down (VTX sheet), no external part -> 0.  
GPIO5 (34) VTX\_SPI\_CLK; internal pull-up, R81 1k into the RTC6705 CMOS input -> 1 (default timing).  
GPIO12 / MTDI (18): no connect, internal pull-down -> VDD\_SDIO 3.3 V for the in-package flash (VDD\_SDIO eFuse burn optional).  
GPIO15 / MTDI (21): no connect, internal pull-up -> boot log on U0TXD (FC ignores it).  
NOT CONNECTED: flash CMD/IO11 (30) and CLK/IO6 (31); NC pins 25, 35, 36, 44, 45, 47, 48; free IO7/8/9/10 (32, 33, 28, 29; PSRAM on the -02 alternate), IO20 (27), I35 (11), I38 (7), I39 (8).  
ELRS LAYOUT JSON (D53, D81): serialLrx 3, serialTx 1, radio\_sck 25, radio\_mosi 32, radio\_miso 33, radio\_nss 27, radio\_rst 26, radio\_busy 36, radio\_dio1 37, led\_rgb 22, button 0, vtx\_nss 19, vtx\_sck 5, vtx\_mosi 14, vtx\_miso -1, vtx\_amp\_pwm 13, vtx\_amp\_vpd 4, vtx\_amp\_vref 2, "vtx\_miso": -1 is written explicitly: an overlay only replaces keys, so an omitted key keeps the stock 23 = the PICO flash DI, which SPIClass(HSPI).begin() turns into an input (flash cut, crash at VTX init); -1 makes the core attach the HSPI default GPIO12 (no connect here) instead, patched ELRS: GPIO12 VTX\_PWR\_CTL (+3V3\_VTX LDO EN through R84 1k), GPIO34 PA\_NTC (ADC1, read in the isTick window, erratum 3.11).  
VTX LINES: series 1k R80/R81/R82 are drawn on the VTX sheet and placed at the ESP32 end (back-feed limit with +3V3\_VTX off, F21). GPIO13 VTXAMP\_PWM = ELRS vtx\_amp\_pwm (RTC6705 PAOUT1 feed PNP, not the SE5004L bias). PA\_DET is ADC2\_CH0 (not readable with Wi-Fi on; not a flight state).  
+3V3 LOAD: ELRS about 0.07 A in flight, Wi-Fi TX peaks 370 mA (DS Table 19, 802.11b 19.5 dBm) in ELRS Wi-Fi mode only (disarmed, bench); +3V3 peak about 0.48 A with the FC and LED, inside the LP5912 500 mA rating; LDO heat and the COUT sum belong to the POWER sheet (P4-12).  
PLACEMENT: SiP site from floorplan v3 (D55, D61); no 85 C halo rule (D69). Alternate part ESP32-PICO-V3-02 (pins 28/29 unused here, P4-14).



# OpenDrone

|                                                       |                  |           |
|-------------------------------------------------------|------------------|-----------|
| Sheet: /RX/<br>File: rx_esp32_sx1281.kicad_sch        |                  |           |
| <b>Title: ELRS 2.4 GHz RX: ESP32-PICO-V3 + SX1281</b> |                  |           |
| Size: A3                                              | Date: 2026-10-09 | Rev: rev1 |
| KiCad E.D.A. 10.0.6                                   |                  | Id: 11/14 |

[illegible]

U22 LP5907-2.85

3V3\_VTX

C122 1uF

GND

PA\_END

IN

OUT

EN

GND

PA\_VREF

C123 4.7uF

GND

U22 LP5907-2.85

U22 LP5907-2.85 (T1 SNVS7978): HD mode removes PA\_VREF with the rail. EN = PA\_EN straight from ESP32 GPIO2 (strap: internal 45k pull-down + the LP5907 1MOhm EN pull-down), C123 4.7 uF 0402 (> 0.7 of effective); C122 1 uF (about 0.5 of effective) only with the direct C108 => U22 route.

DEVIATIONS, bench-gated: (a) VREF window: 2.85 V  $\pm$  2% = 2.79-2.91 V vs SESS004\_VREF H 2.80-2.90 V and the  $\pm$  2% is specified only for VIN >= VOUT = 1 V (VIN here 3.3 V); vs measures PA\_VREF on several units, cold and hot, (b) HD switch-on with ELRS holding PA\_EN high: +3V3\_VTX collapses before C123 discharges, so VOUT exceeds VIN + 0.5 V (LP5907 abs. max. DS 5.1); by using one body-diode drop (<= 0.7 V, < 1 ms, <= 10 uJ) from C123: vs crosses PA\_VREF and +3V3\_VTX (< 0.5 V). Failbacks: C123 down to 1 uF, or the patched ELRS drops PA\_EN on HD\_EN.

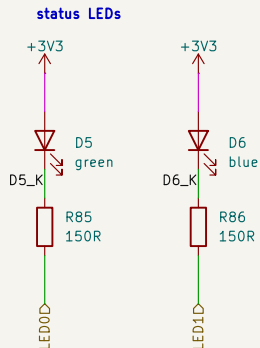
Decoupling at the RTC6705 supply pins (pin numbers next to each capacitor):

**PA DETECTOR + NTC**

The schematic diagram illustrates the PA DETECTOR + NTC circuit. It features a power supply section with +BATT and +5V rails, a PA VCC supply, and a PWR\_FLAG signal. The main circuit includes a SE5004L PA module with pins for VREF, VCC1, VCC2, VCC3, RFout, DET, and NU. The DET pin is connected to a PA\_DET\_RAW signal, which is then filtered by a 1k resistor (R66) and a 100pF capacitor (C17) to produce the PA\_DET signal. The RFout pin is connected to a BPF (FL2) and a UFL (J1). The BPF is a 5.15-5.95G filter, and the UFL is a 5.15-5.95G filter. The PA\_DET\_RAW signal is also connected to a 10k resistor (R70) and a 10k NTC (TH1) to produce the PA\_NTC signal. The circuit is powered by a +5V supply and has a PWR\_FLAG signal input.

**SHEET SOURCE:** Generated by gen\_vtx.py (hardware/tools/sch\_build.py + sch\_report.py). No OpenPDC sibling sheet or common bus exists for an RCTC705 VTX, so the circuit follows the OpenPDC-1-01 reference schematic (RCTC705 supply pins, video network, loop filter, PA0U1 feature), the Skyworks SE5004 data sheet 202393B + EK1 and DESIGN-SPEC 4.B. Res, values, MPNs and footprints: hardware/bom\_plan.json. Signal design: EPS32 controls (RX sheet) on the left, RCTC705 in the middle, PA, BPF and U/LF on the right.

**BENCH ITEMS on this sheet:** RCTC705 pins 11 RF\_VTX, VTX 5spur check; PA Drive Re final value (V5 / V8, 27Rk final value); PALVREF network and the HD switch-on excursion (V5 / V8); PA output (V5, V8) and PA output (V5, V8) and PA output (V5, V8) and PA output (V5, V8) margin (V5; series R 3-5x the 250R ESr, -10°C cold start, slow supply ramp); Q27, R73, R65 temperature at the top reference (V5 / V8, PNP PA DRIVE note).



STATUS LEDs (D5 D6 R85 R86) source: OpenFC-Lite-Mini hardware/rp2350a.kicad\_sch 'status LED' block (R52 + R53 2x 75R, D2 XL-1005UBC), copied and adapted: sink drive per hardware/sch\_contract.json and research/PINMAP.md, 2x 75R -> one 150R 0201, second column for LED1.

Circuit: +3V3 -> LED -> 150R -> GPIO, the pin sinks (contract). Betaflight default polarity, low = on, no LEDx\_INVERTED: LED0\_PIN PA7 (GPIO7, green D5), LED1\_PIN PA26 (GPIO26, blue D6).

Datasheet check (XINGLIGHT XL-1005UGC / UBC, 0402): VF 2.7-3.3 / 2.6-3.4 V at 20 mA, IF max 20 mA, PD 70 mW, VR 5 V. At a few mA the InGaN VF is about 0.2-0.3 V lower (inferred), so with VOL <= 0.5 V at 4 mA drive (RP2350 D5) the LED current is about 0.5-5 mA, typ. 2-3 mA, hard ceiling (3.3 - 2.4) / 150R = 6 mA; LED, 0201 resistor (<= 5.4 mW of 50 mW) and GPIO (4 mA setting, 100 mA IOVSS sum) all inside their ratings; brightness at 1-3 mA is about 17-135 mcd green and 4-36 mcd blue (lv scaled from 350-900 / 80-240 mcd at 20 mA), enough for an indicator. Reverse bias is 0 V when the pin drives high (IOVDD = +3V3).

Reset state: RP2350 pads come up with the pull-down on (PDE = 1, 36-113 kOhm), so both LEDs may glow faintly (about 10-30 uA) until Betaflight drives the pins and in BOOTSEL mode. Accepted, cosmetic.

Placement (P5): bottom side, fc\_core region (bom\_plan); the R may sit at either end of the LED, slow signal (D33). Pins LED0/LED1 = root nets /LED0, /LED1 (RP2350A).

P4 closed here: bom\_plan R85/R86 'P4 checks brightness vs VF' (above). P4-14 fields: RC0201FR-07150RL is a stocked Yageo RC\_L part (Element14 4008725, RS 198-7568: 150R 1 % 50 mW 25 V); no LCSC number known, field left empty (contract open item A9-LCSC).

# OpenDrone

Sheet: /LED/  
File: led.kicad\_sch

**Title: Status LEDs**

Size: A5 Date: 2026-10-09  
KiCad E.D.A. 10.0.6

Rev: rev1  
Id: 13/14

